



Improvements and limitations of the new Climate Hazards Center Infrared Precipitation with Stations (CHIRPSv3) dataset: Insights from multiple spatio-temporal scales in Colombia

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ABSTRACT

The Climate Hazards Center Infrared Precipitation with Stations (CHIRPS) dataset has been widely used in climatological, hydrological, and ecological studies due to its strong agreement with ground-based observations. However, several studies have reported a systematic underestimation of temporal precipitation variability in CHIRPSv2, which is expected to be addressed in the new CHIRPSv3 product. Here, we evaluate the performance of CHIRPSv3 relative to CHIRPSv2 in Colombia, northwestern South America, for the period 2001–2023. We perform point-to-pixel comparisons between ground-based observations from 1004 gauges and both CHIRP (satellite-only product) and CHIRPS (gauge-corrected product) versions 2 and 3 across daily to annual timescales, using continuous, categorical, and extreme precipitation indices. Our results show that CHIRPSv3 outperforms CHIRPSv2, with significantly higher median Kling-Gupta Efficiency (KGE) values (median $KGE_{CHIRPSv3} = 0.20\text{--}0.73$ and $KGE_{CHIRPSv2} = 0.12\text{--}0.67$) across all evaluated temporal and spatial scales. Specifically, CHIRPSv3 provides a better representation of precipitation variability across topographic and climatic gradients, including the wettest regions of the country. These improvements are mainly attributed to a more accurate climatology component (CHPclimv2) that incorporates more ground-based, undercatch-corrected observations, as well as an updated algorithm for estimating the CHIRPv3 product, which more effectively captures a wider range of precipitation values (median $KGE_{CHIRPv3} = 0.14\text{--}0.59$ and $KGE_{CHIRPv2} = 0.09\text{--}0.51$). Despite these improvements, limitations remain in the representation of daily precipitation, particularly in capturing dry-day frequency and precipitation-event intensity. Moreover, extreme precipitation indices describing the duration and intensity of daily precipitation events exhibit low accuracy in Colombia. Nevertheless, our results demonstrate that CHIRPSv3 provides a more accurate representation of spatio-temporal precipitation variability, highlighting its potential for improved climatological, hydrological, and ecological studies.

1. Introduction

Accurate representation of the spatial and temporal variability of precipitation is crucial for climatological, ecological, and hydrological

studies, especially as precipitation events are becoming less frequent but more intense globally under a changing climate (Du et al., 2024; Feldman et al., 2026; Masson-Delmotte et al., 2021). Although ground gauges provide direct precipitation measurements, their records often

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contain missing values, cover short historical periods, and represent a limited area around the instrument, particularly in regions with complex topography (Vargas Godoy et al., 2025). Additionally, their spatial distribution is often irregular, with low densities in regions with limited access, such as oceans, tropical forests, mountainous areas, and deserts (Gebrechorkos et al., 2023; Vallejo-Bernal et al., 2021). This leads to limited representation of the spatio-temporal variability of precipitation, thereby compromising the reliability of decision-making in land and water management (Su et al., 2026).

Gridded precipitation products, such as satellite-based, reanalysis, and multi-source datasets, provide an alternative to overcome the limitations of in situ observations, offering continuous precipitation estimates across a wide range of spatial (0.04° to 2.5°) and temporal (half-hour to annual) resolutions (Sun et al., 2018). Available remote sensing products include the Climate Hazards Center Infrared Precipitation with Stations datasets (CHIRPSv2; Funk et al., 2015), the fifth-generation ECMWF reanalysis for the global climate (ERA5; Hersbach et al., 2020) and the associated land product (ERA5-Land; Muñoz-Sabater et al., 2021), the Integrated Multi-Satellite Retrievals for Global Precipitation Measurement (GPM) Mission (IMERG; Huffman et al., 2020), the NOAA Climate Prediction Center (CPC) MORPHing technique (CMORPHv1; Xie et al., 2017), the Multi-Source Weighted-Ensemble Precipitation (MSWEP; Beck et al., 2019; Wang et al., 2026), and the PERSIANN-Cloud Classification System (PERSIANN-CCS; Nguyen et al., 2018). The accuracy of these products depends on multiple factors, including data sources (e.g., ground observations, satellite, and reanalysis), precipitation estimation methods (i.e., algorithms), temporal resolution, and regional factors such as climatic and topographic features (Mekonnen et al., 2023; Shen et al., 2020; Sun et al., 2018). Thus, numerous evaluations of gridded precipitation products at regional, national, and global scales have been conducted to assess their performance, determine their suitability for specific applications, and identify key limitations to inform future improvements (Du et al., 2024; Pradhan et al., 2022; Shen et al., 2020; Vargas Godoy et al., 2025).

The Climate Hazards Center Infrared Precipitation with Stations (CHIRPSv2) product, developed by the UC Santa Barbara Climate Hazards Center, shows strong agreement with ground-based observations around the world, including regions such as Colombia (Valencia et al., 2023a); Brazil (Baez-Villanueva et al., 2018; Cavalcante et al., 2020; Paredes-Trejo et al., 2017), Eastern Africa (Dinku et al., 2018; Mekonnen et al., 2023), and Southeast Asia (Xie et al., 2025). As a result, it has been widely used for drought monitoring (Harrison et al., 2019; Peng et al., 2020), climate variability and climate change assessments (Arias et al., 2024), ecological and biogeographical studies (Ribeiro et al., 2024; Valencia et al., 2023b), landslide modeling (Gómez et al., 2023), and hydrological modeling (Du et al., 2024), among other applications. CHIRPSv2 was released in 2015, and an updated version; the Climate Hazards Center Infrared Precipitation with Stations version 3 (CHIRPSv3), was released in early 2025 (Funk et al., 2026). These datasets provide high-resolution ($0.05^\circ \times 0.05^\circ$), quasi-global (50°S – 50°N in v2 and 60°S – 60°N in v3, over land only) daily, pentad, and monthly precipitation estimates from 1981 to the present. CHIRPS is developed by integrating satellite information, reanalysis data, and in situ observations from multiple global, national, and regional meteorological networks. This dataset consists of three main components: (i) the Climate Hazards Center Precipitation climatology (CHPclim), (ii) the satellite-only Climate Hazards Center Infrared Precipitation (CHIRP), and (iii) the gauge-corrected CHIRPS product (Funk et al., 2015). CHPclim integrates global monthly historical long-term climatological means from satellite and station observations, along with physiographic indicators, to produce a global 0.05° monthly precipitation climatology. CHIRP is based on a translation of thermal infrared Cold Cloud Duration (CCD) observations from geostationary satellite data into precipitation amounts on a pixel-by-pixel basis. CCD is estimated from the number of hours during which cloud-top brightness temperatures are below 235 K – the most widely used CCD threshold (Funk et al., 2026). The CHIRP

data used in v2 and v3 use different methods for this translation, with CHIRPv3 using a relationship between historical CCD and the CHPclimv2 that is calculated directly (Funk et al., 2022). Conversely, CHIRPv2 additionally incorporated a linear regression with Tropical Rainfall Measuring Mission Multi-satellite Precipitation Analysis version 7 (TMPA 3B42 v7) data. CHIRP has a two-day latency and provides estimates for the most recent pentad, from the 1st, 6th, 11th, 16th, 21st, and 26th days of each month. To create CHIRPS, station observations are blended using a modified inverse-distance weighting procedure. A preliminary version of CHIRPS is generated at the same two-day latency as CHIRP, using a subset of station data, and a final version of CHIRPS, with many more station observations, is generated monthly with a two-to-three-week latency. According to the Climate Hazards Center, from CHIRPSv2 to CHIRPSv3, the CHPclim and CHIRP components were updated to incorporate four major improvements: (i) the inclusion of thousands of additional station observations, (ii) a switch to gauge-undercatch corrected precipitation estimation to account for the wind-effect bias on ground observations proposed by Legates and Willmott (1990), (iii) the incorporation of new, more accurate remote sensing precipitation inputs (TRMM-TMPA 3B42v7 in v2 and IMERGv6 in v3), and (iv) adjustments to the CHIRP algorithm to better estimate a wider range of precipitation values (Funk et al., 2026).

The new CHIRPSv3 product is expected to overcome reported limitations in CHIRPSv2, specifically the tendency to underestimate precipitation magnitude and variance as well as the limited capability to accurately distinguish between daily precipitation intensity classes (e.g., < 1 mm, 1–5 mm, 5–20 mm, among others; Funk et al., 2020, 2026; Mekonnen et al., 2023; Shen et al., 2020). For instance, CHIRPSv2 tends to underestimate precipitation amounts in regions like northern and northeastern Brazil (Cavalcante et al., 2020; Paca et al., 2020; Paredes-Trejo et al., 2017), northern Chile (Zambrano-Bigiarini et al., 2017), Southeast Asia (Alejo and Alejandro, 2021; Xie et al., 2025), Tanzania (Mulungu and Mukama, 2023); and eastern portions of East Africa, North Africa, and the Arabian Peninsula (Funk et al., 2020). These studies show that the largest underestimations occur for the highest daily and monthly precipitation values recorded in ground gauges. Meanwhile, precipitation overestimations are observed across central Africa (Mekonnen et al., 2023; Nicholson and Klotter, 2021), Malaysia (Ayoub et al., 2020; Sa'adi et al., 2023), central-western Argentina (Rivera et al., 2018), Iran (Saeidizand et al., 2018), and Turkey (Aksu and Akgül, 2020). Multiple studies have also reported mixed results with respect to accuracy in different regions of complex topography (e.g.; Baez-Villanueva et al., 2018; Mulungu and Mukama, 2023; Peña-Guerrero et al., 2022; Rivera et al., 2018; Valencia et al., 2023a; Zambrano-Bigiarini et al., 2017). At daily timescales, while CHIRPSv2 shows good accuracy in capturing dry days or non-precipitation days (daily precipitation below 1 mm), its performance decreases in capturing more intense precipitation events (Ahmed et al., 2024; Baez-Villanueva et al., 2018; Harrison et al., 2019; Mekonnen et al., 2023; Peña-Guerrero et al., 2022; Zambrano-Bigiarini et al., 2017).

Accurately estimating precipitation from remote sensing remains challenging in northwestern South America due to extensive ungauged areas (Su et al., 2026), large mesoscale convective systems (Feng et al., 2021; Hernandez-Deckers, 2022), and complex mountainous terrain, with Colombia presenting particular difficulties because of its large topographic and climatic gradients (Poveda et al., 2025; Sun et al., 2018; Fig. 1). This territory, therefore, provides ideal conditions to evaluate and compare the performance of these products (Vallejo-Bernal et al., 2021). Previous studies have identified CHIRPSv2 as one of the most accurate datasets across multiple spatial and temporal scales in Colombia (Baez-Villanueva et al., 2018; Cepeda Arias and Cañon Bariga, 2022; López-Bermeo et al., 2022; Ocampo-Marulanda et al., 2022). However, these studies also reported that CHIRPSv2 tends to both underestimate and overestimate precipitation magnitude depending on the region, while systematically underestimating its temporal variability at daily and monthly scales across the country. Specifically, lower

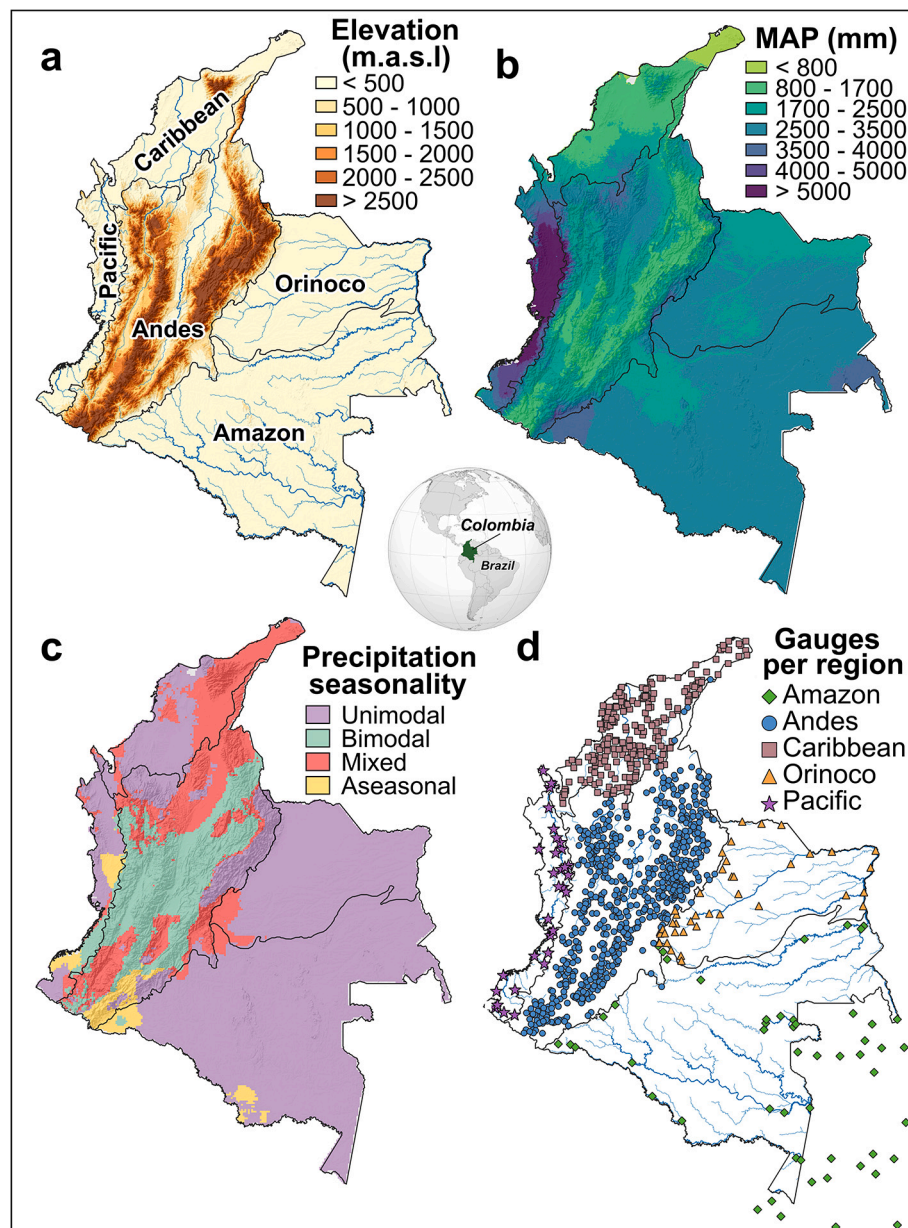


Fig. 1. Topographic and climatic gradients of Colombia and its five major natural regions. (a) Digital elevation model (Jarvis et al., 2008) and the main rivers (Lehner and Grill, 2013), (b) Mean annual precipitation (MAP) from WorldClimv2.1 (Fick and Hijmans, 2017), (c) Precipitation seasonality regimes (Urrea et al., 2019), (d) Location of the 1004 selected precipitation gauges. The natural regions are defined by the Instituto Geográfico Agustín Codazzi (IGAC): Caribbean, Pacific, Andes, Orinoco, and Amazon (IGAC, 1997).

performance is observed for daily-scale measurements in mountainous regions and in the extreme dry and wet regions of the country (Romero-Hernández et al., 2024; Valencia et al., 2023a). Additionally, as observed around the world, CHIRPSv2 is effective at capturing non-precipitation events (< 1 mm/day), but tends to either overestimate or underestimate the frequency of precipitation events (≥ 1 mm/day) in Colombia (Baez-Villanueva et al., 2018; Mekonnen et al., 2023; Valencia et al., 2023a). Therefore, we use Colombia as a testbed to examine the improvements and persistent limitations of the new CHIRPv3 and CHIRPSv3 relative to their previous versions (v2) across multiple temporal scales in Colombia. This is critical for informing future product development and supporting its application across tropical regions.

Here, we assess the performance of CHIRPSv3 relative to CHIRPSv2 at multiple temporal scales across Colombia and its five natural regions for the period 2001–2023. First, we perform point-to-pixel comparisons between ground-based observations from 1004 gauges and both CHIRP

(satellite-only product) and CHIRPS (gauge-corrected product) versions 2 and 3 across daily to annual timescales using the Kling-Gupta Efficiency. Next, using categorical indices, we assess the ability of CHIRPSv2 and CHIRPSv3 to reproduce the occurrence of different precipitation intensity categories. Finally, we evaluate the differences between CHIRPSv2 and CHIRPSv3 in capturing extreme precipitation indices used to characterize the amount, duration, frequency, and intensity of extreme precipitation events across Colombia.

2. Data and methods

2.1. Study area

Located in northwestern South America, the continental portion of Colombia spans approximately $1'100'000$ km². It is characterized by pronounced topographic and climatic gradients representative of those

found across the tropics (Fig. 1). The country exhibits complex topography due to the branching of the Andes into three mountain ranges that extend northward, with altitudes ranging from 0 to 5667 m.a.s.l. (Fig. 1a). This topographical setting, combined with multiple and interacting atmospheric drivers – such as the Intertropical Convergence Zone (ITCZ), regional low-level jets, and the Amazon–Andes connection – result in a wide variety of climatic conditions and landscapes, including deserts and xeric shrublands, savannas, tropical forests, and high-mountain ecosystems (e.g., *páramos*) (Escobar et al., 2022; Espinoza et al., 2020; Poveda et al., 2025). Mean annual precipitation (MAP) varies from more than 5000 mm on the Pacific coast to less than 300 mm in northern arid zones (Fig. 1b). Similarly, seasonal precipitation regimes exhibit multiple patterns, including unimodal, bimodal, mixed, and aseasonal (i.e., with no marked seasonality) (Urrea et al., 2019) (Fig. 1c). Therefore, the country is divided into five natural regions based on ecosystems, topographic complexity, and climate, as defined by the Instituto Geográfico Agustín Codazzi (IGAC): Caribbean, Pacific, Andes, Orinoco, and Amazon (IGAC, 1997) (Fig. 1a). As in many regions globally, precipitation gauges in Colombia are predominantly located in easy-to-access areas that include populated, low-elevation areas (< 1500 m.a.s.l.) and along major rivers (Fig. 1d and Fig. S1.1), resulting in limited coverage across remote (e.g., Amazon and Orinoco basins) and high-elevation areas (> 3000 m.a.s.l.). Multiple studies have assessed the performance of spatially continuous, gridded precipitation products (e.g., CHIRPSv2, CMORPHv1, ERA5, ERA5-Land, IMERG, MSWEPv2.8, and PERSIANN-CCS) in Colombia, indicating that CHIRPSv2 shows higher performance relative to other products at both regional and national scales, from daily to annual timescales (Baez-Villanueva et al., 2018; Cepeda Arias and Cañon Barriga, 2022; López-Bermeo et al., 2022; Ocampo-Marulanda et al., 2022; Romero-Hernández et al., 2024; Valencia et al., 2023a).

2.2. Ground-based precipitation observations

We obtained daily precipitation records from 1634 gauges over the 1980–2024 period from the Colombian Institute of Hydrology, Meteorology, and Environmental Studies (IDEAM). We selected January 1, 2001, to December 31, 2023, as our study period to obtain the largest number of gauges, the lowest percentage of missing values (<10%), and the best spatial coverage. This is important because the spatial distribution of gauges in Colombia is highly non-uniform, with most gauges concentrated in the lower-elevation areas of the Andes and Caribbean regions (Fig. 1d and Fig. S1.1). For this reason, we selected a missing-values threshold higher than those used in previous studies (e.g., 2% for 2003–2010 in Zambrano-Bigiarini et al. (2017) and 5% for 2000–2016 in Baez-Villanueva et al. (2018)) as a compromise between temporal and spatial coverage. However, most of the selected gauges (~63%) have less than 5% of missing values (Fig. S1.1c). Specifically, we defined missing values as days with no recorded measurements or precipitation observations flagged as preliminary data (i.e., subject to changes or modifications according to IDEAM methods) that exceeded four standard deviations from each gauge's mean. To improve the spatial coverage of ground gauges in the Amazon region, we also included 24 gauges in neighboring Brazil, located along the Colombia border, using data from the Brazilian Daily Weather Gridded Dataset (Xavier et al., 2022). We again used data from the 2001–2023 period and applied the same threshold for low-quality data and missing values. As a result, we included 1004 gauges in our analysis, located at elevations between 1 and 3650 m.a.s.l., and MAP levels between 225 and 12400 mm (Fig. S1.1d).

2.3. Updated CHIRP and CHIRPS products

The development of the updated CHIRPv3 and CHIRPSv3 products included a new, more reliable CHPclim dataset (CHPclimv2). CHPclimv2 follows a similar approach to CHPclimv1 but incorporates

IMERG inputs for improving precipitation estimates and adds ~84000 new gauge-undercatch corrected station normals from the Global Precipitation Climatology Center (GPCC) to address systematic biases, resulting in an overall higher mean precipitation compared to CHPclimv1 (see Figs. S1.3–S1.4). To address gauge-undercatch in ground-gauge precipitation, a monthly correction factor proposed by Legates and Willmott (1990) is applied when blending stations to estimate CHIRPSv3. This correction accounts for the impact of wind on the amount of precipitation captured by gauges, especially in mountainous regions – an effect not incorporated in CHIRPSv2 (see Fig. S1.5). The new CHIRPv3 algorithm also estimates precipitation from geostationary satellite thermal infrared measurements of CCD, but version 3.0 produces a wider range of values, including more zero values (instead of near-zero values) and higher high-range values in most areas. This can be seen in the larger standard deviations of precipitation estimates in CHIRPv3 and CHIRPSv3 relative to CHIRPv2 and CHIRPSv2, respectively (Funk et al., 2026). While CHIRPSv3 maintains the low mean bias of CHIRPSv2, and both products tend to be highly correlated, the wider range of precipitation in v3 can lead to higher accuracy (see Figs. S2.1–S2.2). Additionally, pentad CHIRPSv3 estimates are disaggregated to a daily scale using daily IMERG late (CHIRPSv3-IMERG) and ERA5 (CHIRPv3-ERA5), while daily CHIRPSv2 estimates are based on daily NCEP Climate Forecast System data (Funk et al., 2015, 2026).

2.4. Data comparison

We applied a point-to-pixel analysis to compare the precipitation time series of each selected gauge with the corresponding CHIRP and CHIRPS pixel using multiple temporal scales: daily (excluding CHIRP), pentad (5-day), monthly, seasonal (dry and wet seasons), and annual. These temporal scales were selected based on their widespread use in previous validation studies (e.g., Alejo and Alejandro, 2021; Baez-Villanueva et al., 2018; Cavalcante et al., 2020; Dinku et al., 2018; Valencia et al., 2023a) and their relevance for CHIRPS applications, including early warning and hydrological modeling (daily), crop and drought monitoring (pentad and monthly), and climatological studies (seasonal and annual). Daily ground-based observations from each gauge were aggregated into pentads, monthly, seasonal, and annual scales. For the pentad analysis, any period with at least one missing value was excluded. For monthly, seasonal, and annual comparisons, periods were excluded if they contained more than 5, 15, or 30 missing daily observations, respectively. Due to the diverse precipitation regimes across the country (Fig. 1c), we defined the dry and wet seasons in each gauge as the three driest and wettest months, respectively, based on the mean climatology for the 2001–2023 period.

Many ground observations in Colombia were incorporated into the development of CHIRPSv2 and CHIRPSv3, with the newer version including more observations (Fig. S1.2). To avoid confusion between the gauges used to produce CHIRPS and those used to evaluate their performance in Colombia (i.e., non-independent and independent gauges of CHIRPS), we refer to the former as “stations” throughout the manuscript, while we refer to the latter as “gauges”. Because it is challenging to avoid overlapping between stations and gauges as independent gauges vary monthly due to data availability and quality control, we examined how the performance of both CHIRPS versions varies with station density. Specifically, we extracted the number of ground stations blended in CHIRPSv2 and CHIRPSv3 per 0.25° pixel and calculated the mean number of ground stations over the 2001–2023 period. Ultimately, we also extracted the CHPclimv1 and CHPclimv2 data (Figs. S1.3–1.4), along with gridded monthly gauge-undercatch correction factors used to adjust CHPclimv2 and CHIRPSv3 (Fig. S1.5). The monthly wind-bias correction factors were used to compare CHIRPSv3 estimates relative to undercatch-corrected and raw – uncorrected – precipitation observations from 1004 selected gauges at daily, pentad, and monthly timescales. This is important because the CHIRPS developers – the Climate Hazards Center, UC Santa Barbara – recommend

using gauge-corrected data when comparing CHIRPSv3 with ground observations (Funk et al., 2026).

2.5. Performance indices

2.5.1. Continuous and categorical indices

We examined the performance of CHIRP and CHIRPS (versions 2 and 3) at pentad, monthly, seasonal, and annual scales, as well as CHIRPS at the daily scale, using multiple continuous and categorical indices. The continuous index included the modified Kling-Gupta Efficiency (KGE, Gupta et al., 2009; Kling et al., 2012), which decomposes the performance into linear correlation, bias ratio, and variability ratio, as described in Table 1. The optimal value of KGE and its three components is 1. The Pearson correlation coefficient (r) measures the linear correlation between precipitation observations and values estimated by CHIRP or CHIRPS, ranging from -1 (perfect negative correlation) to 1 (perfect positive correlation), with 0 indicating no correlation. The bias ratio (β) indicates the average tendency of CHIRP or CHIRPS to estimate precipitation observations; such that $\beta > 1$ indicates that precipitation estimates overestimate the mean ground observations, and $\beta < 1$ indicates underestimation of the mean observed values. The variability ratio (γ) evaluates whether the dispersion of CHIRP or CHIRPS precipitation estimations is greater ($\gamma > 1$) or smaller ($\gamma < 1$) than that of the observations. Therefore, while KGE indicates the overall performance of CHIRP or CHIRPS, its components reflect how well the precipitation estimates capture the temporal correlation, magnitude, and variability of ground observations. Finally, we assessed the statistical significance differences in the performance of CHIRP and CHIRPS versions 2 and 3 using a two-sided nonparametric Wilcoxon signed-rank test (Woolson, 2008).

We also used categorical indices to assess the performance of CHIRPS at different daily precipitation intensity levels as defined by Baez-Villanueva et al. (2018) and Valencia et al. (2023a) in previous analyses in Colombia: (i) non-precipitation (dry days; < 1 mm/day), (ii) light precipitation [1 – 5] mm/day, (iii) moderate precipitation [5 – 20] mm/day, (iv) heavy precipitation [20 – 40] mm/day, and (v) violent precipitation

(≥ 40 mm/day). Categorical indices included the probability of detection (POD), the false alarm ratio (FAR), the Hansen–Kuipers discriminant (HK), and the frequency bias (fBIAS) (Table 1). The POD and FAR indices measure the fraction of events that are correctly and incorrectly identified by CHIRPS, respectively. Both indices range from 0 to 1 , with 1 being the optimum value for POD while 0 is the optimum value for FAR (i.e., non-events are incorrectly identified). The HK, also known as the True Skill Statistic (TSS), measures CHIRPS's discrimination ability in capturing events and non-events, exhibiting low sensitivity to imbalances in their frequencies. HK ranges from -1 to 1 , with 1 being its optimal value, scores below 0 indicating that the chance forecast of the event should be preferred over the actual unskilled CHIRPS value. The fBIAS compares the ratio of events identified by CHIRPS to the number of events reported by ground observations. The optimal value of fBIAS is 1 (unbiased), meaning that the number of events registered by CHIRPS is equal to the number of events registered by the precipitation gauge. fBIAS > 1 indicates an overestimation of the occurrences by CHIRPS, whereas fBIAS < 1 reveals the opposite. All data processing and analyses were conducted using R version 4.3.2 (R Core Team, 2023).

2.5.2. Extreme precipitation indices

The new CHIRPSv3 algorithm is expected to improve the estimation of a wider range of precipitation values compared to CHIRPSv2, particularly in capturing large precipitation amounts during high-impact, off-season, or extreme storm events (Funk et al., 2026). Therefore, we also assessed the capability of CHIRPSv3, both CHIRPSv3-ERA5 and CHIRPSv3-IMERG, relative to CHIRPSv2 in identifying extreme precipitation events, in terms of amount, duration; frequency, and intensity, using ten extreme precipitation indices recommended by the Expert Team on Climate Change Detection and Indices (ETCDDI; Zhang et al., 2011). The amount category indices include the total annual precipitation during wet days (PRCPTOT), very wet days (R95p), and extreme wet days (R99p). The maximum annual number of consecutive dry (CDD) and wet (CWD) days are used to characterize the duration category. For the frequency category, the indices are the number of days with daily precipitation above 10 mm (R10mm) and 20 mm (R20mm).

Table 1

Description of continuous and categorical indices used in this study, including their equation, range, and optimal value.

Metrics	Equation		Range	Optimal value
<i>Continuous indices</i>				
Modified Kling-Gupta Efficiency	$KGE = 1 - \sqrt{(r-1)^2 + (\beta-1)^2 + (\gamma-1)^2}$	(1)	$(-\infty, 1]$	1
Pearson correlation coefficient	$r = \frac{\sum_{i=1}^n (O_i - \bar{O})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (O_i - \bar{O})^2} \sqrt{\sum_{i=1}^n (S_i - \bar{S})^2}}$	(2)	$[-1, 1]$	1
Bias ratio	$\beta = \frac{\bar{S}}{\bar{O}}$	(3)	$(-\infty, +\infty)$	1
Variability ratio	$\gamma = \frac{CV_s}{CV_o} = \frac{\sigma_s/\bar{S}}{\sigma_o/\bar{O}}$	(4)	$(-\infty, +\infty)$	1
<i>Categorical indices</i>				
Probability of detection	$POD = \frac{H}{H+M}$	(5)	$[0, 1]$	1
False alarm rate	$FAR = \frac{FA}{H+FA}$	(6)	$[0, 1]$	0
Hansen-Kuipers skill score	$HK = \frac{H}{H+M} - \frac{FA}{FA+CN}$	(7)	$[-1, 1]$	1
Frequency bias	$fBIAS = \frac{H+FA}{H+M}$	(8)	$[0, +\infty)$	1
<i>Performance metrics for extreme precipitation indices</i>				
Median percent bias	$mBIAS = 100 \times \frac{\text{median}(S_i) - \text{median}(O_i)}{\text{median}(O_i)}$	(9)	$(-\infty, +\infty)$	0
Spearman's rank correlation	$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$	(10)	$[-1, 1]$	1

Notation: n number of observations, O_i and S_i are the observed (precipitation gauges) and the corresponding CHIRP or CHIRPS values for day, month, or year i . $\bar{O}$ and $\bar{S}$ are the mean observed and estimated precipitation values. H indicates a hit, i.e., CHIRPS correctly identifies the type of precipitation event measured at the gauge, M is a miss, i.e., an event recorded at the precipitation gauge but not correctly identified by CHIRPS, FA is a false alarm, i.e., a precipitation event detected by CHIRPS but not recorded at the corresponding precipitation gauge, CN represents the correct negatives (or correct rejections), i.e., an event not registered either by the precipitation gauge or CHIRPS. Table S1 summarizes the contingency table used to compute the different categorical indices.

The intensity category indices include the annual maximum 1-day (Rx1day) and consecutive 5-day precipitation (Rx5day), as well as the mean daily intensity of wet days (SDII; see details in Table 2). The performance of CHIRPSv2, CHIRPSv3-IMERG, and CHIRPv3-ERA5 in capturing each extreme precipitation index was evaluated based on the median percent bias and Spearman's rank correlation (Table 1) to appropriately assess continuous (PRCPTOT, Rx1day, Rx5day, SDII), threshold-based count indices (R95p, R99p, R10mm, R20mm), and sequence-dependent indices (CDD, CWD). The mBIAS quantifies the tendency of CHIRPS to overestimate (mBIAS > 0) or underestimate (mBIAS < 0) median ground observations, with an optimal value of 0 (unbiased). The Spearman's rank correlation (ρ) measures the strength and direction of the relationship between ranked CHIRPS and ground-based observations.

3. Results

3.1. Performance from pentad to annual scales

CHIRPSv3 outperforms CHIRPSv2 at pentad, monthly, seasonal (both dry and wet seasons), and annual scales, both at the country scale (Fig. 2 and Figs. S2.1–2.2) and across the five natural regions (Figs. S2.3–S2.7), as reflected by significantly higher median KGE values ($p < 0.05$). Similarly, the satellite-only CHIRPv3 product performs significantly better than CHIRPv2. The highest performance is observed at the monthly scale, followed by pentad, seasonal, and annual scales, with CHIRPS, as expected, showing better results than CHIRP in both versions due to station-based bias correction. The systematic

Table 2

Description of the extreme precipitation indices used in this study, as recommended by the Expert Team on Climate Change Detection and Indices (ETCCDI; Zhang et al., 2011).

Category	Index	Indicator	Definitions	Units
Amount	PRCPTOT	Annual total wet-day precipitation	Total annual precipitation from days with daily precipitation ($P \geq 1$ mm)	mm
	R95p	Very wet days	Total annual precipitation from days >95th percentile of days with $P \geq 1$ mm	mm
	R99p	Extreme wet days	Total annual precipitation from days >99th percentile of days with $P \geq 1$ mm	mm
Duration	CDD	Consecutive dry days	Maximum number of consecutive days when $P \geq 1$ mm	days
	CWD	Consecutive wet days	Maximum number of consecutive days when $P < 1$ mm	days
Frequency	R10mm	Count of days when precipitation ≥ 10 mm	Number of days with $P \geq 10$ mm	days
	R20mm	Count of days when precipitation ≥ 20 mm	Number of days with $P \geq 20$ mm	days
	Rx1day	Max 1-day precipitation amount	Annual maximum 1-day precipitation	mm
Intensity	Rx5day	Max 5-day precipitation amount	Annual maximum consecutive 5-day precipitation	mm
	SDII	Simple daily intensity index	The ratio of annual total precipitation to the number of wet days ($P \geq 1$ mm)	mm/day

improvements in KGE values for both CHIRPv3 and CHIRPSv3 are mainly associated with slight increases in the correlation coefficient (r), a narrower distribution of bias ratio (β), and lower underestimation of precipitation variability (γ). For example, both versions of CHIRP and CHIRPS products are almost unbiased ($\beta = 1$), but the new variants (v3) exhibit distributions more centered around the optimal value at the pentad, monthly, and annual scales. This suggests that CHIRPv3 and CHIRPSv3 tend neither to overestimate nor underestimate the mean precipitation magnitude. Additionally, the median variability ratio also shows a lower underestimation of precipitation variability at the pentad scale ($p < 0.05$), increasing from 0.70 to 0.76 for CHIRPS (v2 to v3) and from 0.60 to 0.75 for CHIRP (v2 to v3). At monthly and annual scales, the underestimation of precipitation variability also decreases, but the changes are smaller than those observed at the pentad scale. Furthermore, our results indicate that the most pronounced performance differences between CHIRP and CHIRPS versions 2 and 3 are reflected in the correlation coefficient and variability ratio, whereas the bias ratio remains relatively similar between v2 and v3. Finally, both CHIRPv3 and CHIRPSv3 products exhibit stronger linear correlations and lower root mean square error (RMSE) compared to observations than CHIRPv2 and CHIRPSv2. However, they tend to show higher relative bias (BIAS), with a shift from precipitation underestimations (BIAS < 0) to overestimations (BIAS > 0) at pentad, monthly, seasonal (both dry and wet seasons), and annual scales (Figs. S2.1–2.2). Importantly, our results suggest that CHIRP and CHIRPS do not show systematic performance differences between Colombian and Brazilian gauges, despite potential differences in instrumentation and operational protocols.

Overall, these country-scale patterns remain consistent at the regional scale (Figs. S2.3 to S2.7). However, the performance of both CHIRP and CHIRPS products – versions 2 and 3 – decreases across all temporal scales in the Amazon and Pacific regions, which are the wettest areas of the country and the most poorly sampled with gauges (Figs. S2.3–S2.4). The lower KGE values are mainly due to lower correlation coefficients and greater underestimations of precipitation variability. Conversely, our results show better performance in the Orinoco region compared to the national scale for both CHIRP and CHIRPS versions. This improvement is primarily driven by higher temporal correlations and lower underestimations of precipitation variability (γ), particularly at the pentad and monthly scales (Fig. S2.5).

Our results also show that CHIRPv3 and CHIRPSv3 perform better than CHIRPv2 and CHIRPSv2, respectively, and that their estimation error exhibits weaker dependence on topographic and climatic gradients across Colombia at pentad and monthly scales (Figs. S3.1–3.2). Indeed, correlation coefficients are slightly higher in version 3 than in version 2 for both CHIRP and CHIRPS products along topographic (0 to 3750 m.a.s.l.) and MAP (225 to 12,400 mm) gradients. The results indicate that both CHIRPv2 and CHIRPSv2 exhibit an elevation-dependent bias, characterized by overestimation of precipitation ($\beta > 1$) at low elevations (< 1000 m.a.s.l.), underestimation ($\beta < 1$) at mid-elevations (1000–2500 m.a.s.l.), and overestimation at higher elevations (> 2500 m.a.s.l.). Conversely, CHIRPv3 and CHIRPSv3 exhibit slight overestimations that are consistent across all elevations. Regarding, MAP gradients, CHIRPv2 and CHIRPSv2 tend to overestimate precipitation in regions with MAP between 500 and 1500 mm, while both products underestimate precipitation in wetter areas, particularly where MAP exceeds 3500 mm, consistent with the patterns shown in Figs. S2.1–2.2. In contrast, CHIRPv3 and CHIRPSv3 exhibit smaller and less variable overestimations across MAP gradients, except in drier regions (MAP < 1000 mm), where overestimations are more pronounced. Similarly, the new versions of CHIRP and CHIRPS (v3) show smaller underestimations of precipitation variability than their previous version (v2) along topographic and MAP gradients.

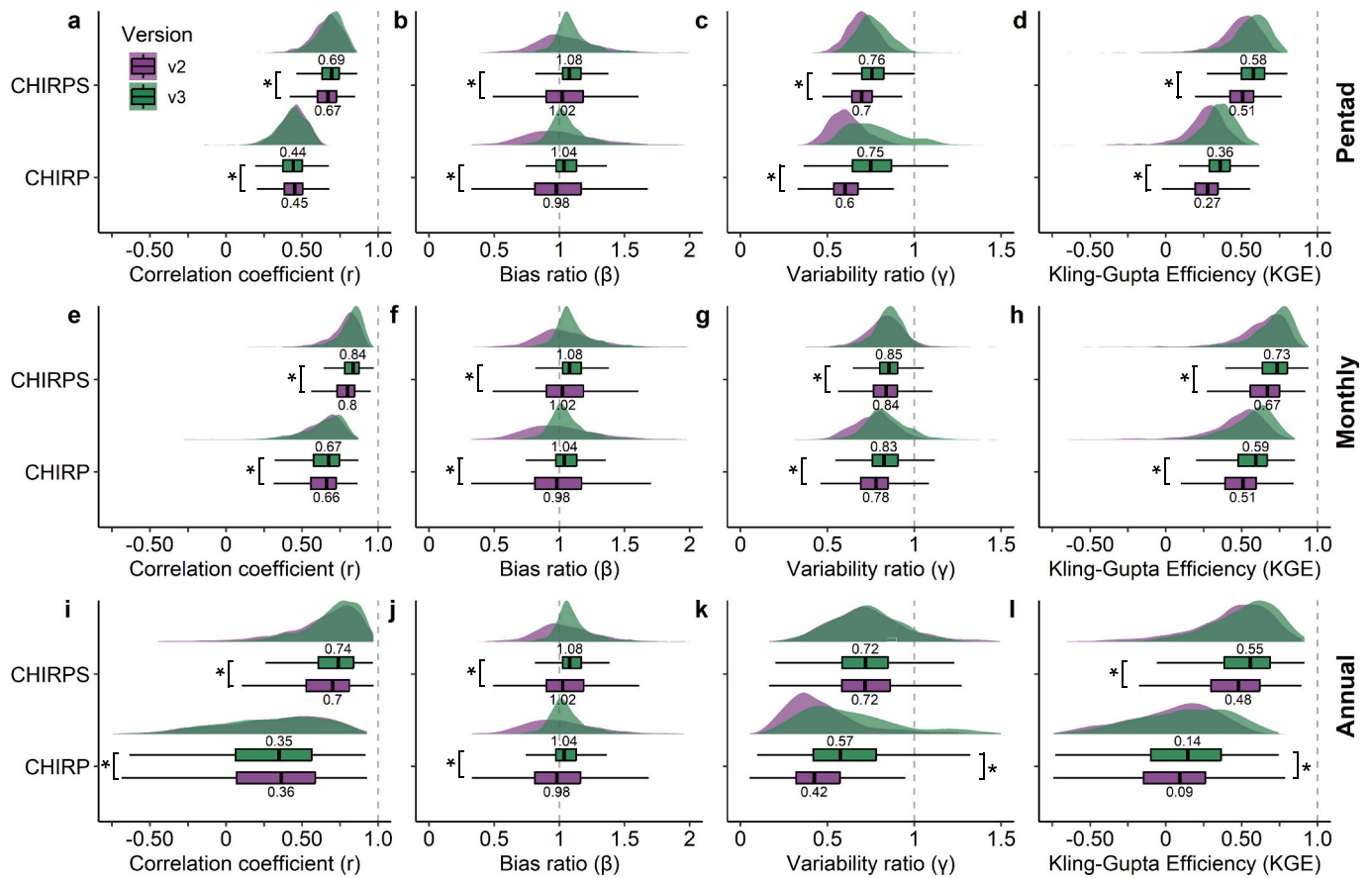


Fig. 2. Performance of CHIRP and CHIRPS versions 2 (v2) and 3 (v3) at the (a–d) pentad, (e–h) monthly, and (i–l) annual scales, as indicated by the Modified Kling-Gupta Efficiency (KGE), and its three components: correlation coefficient (r), bias ratio (β), and variability (γ) ratio. The dashed vertical grey line in all panels indicates the optimal value for each index. Median values for each product and metric are shown above or below the corresponding boxplots. Asterisks indicate statistically significant difference ($p < 0.05$) between CHIRPSv3 and CHIRPSv2, based on a two-sided nonparametric Wilcoxon signed-rank test.

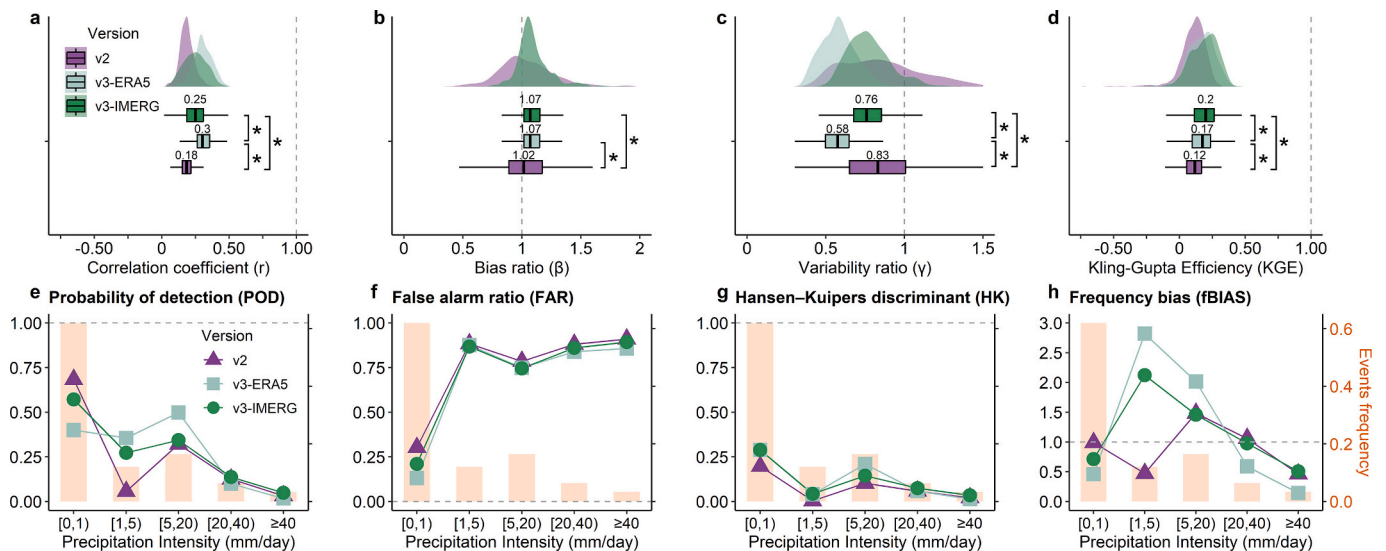


Fig. 3. (a–d) Performance of CHIRPSv2 (v2), CHIRPSv3-ERA5 (v3-ERA5), and CHIRPSv3-IMERG (v3-IMERG) at the daily scale, evaluated using the Modified Kling-Gupta Efficiency (KGE) and its three components: correlation coefficient (r), bias ratio (β), and variability ratio (γ). Median values for each product and metric are shown above the corresponding boxplot. Asterisks indicate statistically significant difference ($p < 0.05$) between CHIRPS products (e.g., CHIRPSv3-IMERG and CHIRPSv2 or CHIRPSv3-IMERG and CHIRPSv3-ERA5), based on a two-sided nonparametric Wilcoxon signed-rank test. (e–h) Median values of categorical performance indices for CHIRPSv2 (triangle), CHIRPSv3-ERA5 (square), and CHIRPSv3-IMERG (circle) across five daily precipitation intensity levels. Non-precipitation (< 1 mm), light precipitation ([1–5] mm), moderate precipitation ([5–20] mm), heavy precipitation ([20–40] mm), and violent precipitation (≥ 40 mm). Bars represent the fraction of daily observed precipitation events for each intensity category. The grey dashed line in all panels indicates the optimal value for each index.

3.2. Performance at the daily scale

3.2.1. Continuous and categorical indices

Daily CHIRPSv3 products (v3-ERA5 and v3-IMERG) significantly outperform CHIRPSv2 across Colombia, with CHIRPSv3-IMERG showing the highest overall KGE values ($p < 0.05$; Figs. 3a–3d). Both CHIRPSv3-based products exhibit higher correlation coefficients (r) with ground gauges than CHIRPSv2, mainly for CHIRPSv3-ERA5 (Fig. 3a). As for the results across temporal scales from pentad to annual (Fig. 2), all three products are nearly unbiased ($\beta \approx 1$), but CHIRPSv3-ERA5 and CHIRPSv3-IMERG exhibit a narrower distribution of bias ratios compared to CHIRPSv2 (Fig. 3b). However, CHIRPSv3-ERA5 ($\gamma = 0.58$) and CHIRPSv3-IMERG ($\gamma = 0.76$) show a higher underestimation of precipitation variability than CHIRPSv2 ($\gamma = 0.83$) at daily scale (Fig. 3c). Although these patterns generally remain consistent when each natural region is considered individually, the performance of the CHIRPS products improves in the Orinoco region and declines in the Caribbean (the driest) and Pacific (the wettest) regions compared to the country-scale results (Figs. S4.1–4.5). Moreover, our results show consistently improved performance of both CHIRPSv3-ERA5 and CHIRPSv3-IMERG products relative to CHIRPSv2 across the topographic (Fig. S4.6) and MAP (Fig. S4.7) gradients at the daily scale, as described at the pentad and monthly scales (Figs. S3.1–3.2).

The results of the categorical indices indicate that the ability of the CHIRPS products in capturing precipitation varies across daily event categories and among the different products – CHIRPSv2, CHIRPSv3-ERA5, and CHIRPSv3-IMERG (Figs. 3e–3h). Overall, the CHIRPS products exhibit a higher probability of detection (POD) for non-precipitation events (< 1 mm/day) – the most frequent events – particularly for CHIRPSv2 (POD = 0.68) and CHIRPSv3-IMERG (POD = 0.57) versions (Fig. 3e). Both CHIRPSv3-based products improve the POD for light ([1–5] mm/day) and moderate ([5–20] mm/day) precipitation events compared to CHIRPSv2. However, similar POD values are observed for heavy ([20–40] mm/day) and violent (≥ 40 mm/day) precipitation events among CHIRPSv2, CHIRPSv3-ERA5, and CHIRPSv3-IMERG. Similarly, CHIRPS products exhibit low false alarm rates (FAR) for non-precipitation events (FAR < 0.30), but higher FAR values for all other precipitation intensities (Fig. 3f). In particular, CHIRPSv3-ERA5 (FAR = 0.13) and CHIRPSv3-IMERG (FAR = 0.21) exhibit lower FAR values than CHIRPSv2 (FAR = 0.30) for non-precipitation events. However, there is no difference among CHIRPS products for events with precipitation intensities above 1 mm/day. The Hansen-Kuipers discriminant (HK) shows that all CHIRPS products exhibit a low discrimination skill (HK < 0.3), particularly for light, heavy, and violent precipitation events (Fig. 3g). However, CHIRPSv3-IMERG and CHIRPSv3-ERA5 slightly outperform CHIRPSv2, particularly for non-precipitation and moderate precipitation events. Finally, the frequency bias (fBIAS) indicates that CHIRPS products tend to either underestimate (fBIAS < 1) or overestimate (fBIAS > 1) the number of precipitation events, depending on their intensity (Fig. 3h). For non-precipitation events, CHIRPSv2 is nearly unbiased (fBIAS ≈ 1), whereas both CHIRPSv3-ERA5 (fBIAS = 0.46) and CHIRPSv3-IMERG (fBIAS = 0.71) underestimate the number of events. Interestingly, while CHIRPSv2 underestimates light precipitation events (fBIAS = 0.47), both CHIRPSv3 products largely overestimate these events (fBIAS > 2.12), particularly CHIRPSv3-ERA5 (fBIAS = 2.82). For moderate, heavy, and violent precipitation events, the CHIRPS products exhibit both overestimations and underestimations, with CHIRPSv2 and CHIRPSv3-IMERG showing comparable performance, whereas CHIRPSv3-ERA5 exhibits a larger bias.

Because non-precipitation and light events are more frequent during the dry season compared to the wet season, we also examined the daily categorical indices separately for each season (Fig. S4.8). During the dry season, the three CHIRPS products more accurately capture non-precipitation events than during the wet season and full year data, in terms of probability of detection (POD > 0.6), false alarm ratio

(FAR < 0.25), Hansen-Kuipers discrimination (HK > 0.15), and frequency bias (fBIAS ≈ 1). Conversely, CHIRPSv3-ERA5 and CHIRPSv3-IMERG exhibit a decrease in their ability to capture non-precipitation events during the wet season. Interestingly, the overestimations in the number of light precipitation events ([1, 5] mm) in CHIRPSv3 products are more pronounced during the dry season than during the wet season, particularly in CHIRPSv3-ERA5.

The categorical indices also indicate regional differences, particularly for non-precipitation and light precipitation events. In the Andes region, the indices show patterns similar to those observed at the national scale, but with lower overestimation of light and moderate precipitation events (Fig. S4.1). The Amazon region similarly exhibits lower fBIAS values for light and moderate precipitation events with CHIRPSv3, however, the results show lower POD and higher FAR values for non-precipitation events across all CHIRPS products compared to the national scale (Fig. S4.2). Conversely, CHIRPSv3 products exhibit a larger overestimation of light events in the Orinoco region (Fig. S4.3). However, all products show improved performance for non-precipitation events in terms of POD, FAR, HK, and fBIAS in this region. In the Pacific region, overestimation and underestimation of light precipitation events increase for CHIRPSv3 products and CHIRPSv2, respectively (Fig. S4.4). Nevertheless, fBIAS improves for moderate, heavy, and violent events, which are more frequent in this region. Additionally, POD decreases and FAR increases for non-precipitation events, like the pattern observed in the Amazon region. Finally, CHIRPS products perform better for non-precipitation events in the Caribbean region than the national median, as indicated by higher POD and HK, and lower FAR and fBIAS values (Fig. S4.5). As in other natural regions, the number of light precipitation events is overestimated in this region, with larger magnitudes for CHIRPSv3-ERA5 (fBIAS = 8.13) and CHIRPSv3-IMERG (fBIAS = 5.18). Light precipitation events are also largely overestimated by CHIRPSv2 (fBIAS = 4.17), in contrast to the results observed in the other natural regions (Fig. S4.9).

3.2.2. Extreme precipitation indices

The performance of both CHIRPSv3-ERA5 and CHIRPSv3-IMERG exhibits contrasting patterns relative to CHIRPSv2 in capturing extreme precipitation indices across Colombia (Fig. 4 and Fig. S4.10). CHIRPSv3-ERA5 and CHIRPSv3-IMERG outperform CHIRPSv2 for PRCPTOT, R95p, R99p, and R10mm, as reflected by lower median percent bias (mBIAS; $p < 0.05$) and higher Spearman's rank correlation ($p < 0.05$). Interestingly, while CHIRPSv3-ERA5 is almost unbiased (median mBIAS $\sim 1\%$), CHIRPSv3-IMERG tends to overestimate (median mBIAS $> 10\%$) the observed R95p, R99p, and R10mm indices. Conversely, both CHIRPSv3 products show greater underestimation and overestimation than CHIRPSv2 for CDD, CWD, Rx1day, and SDII, despite significant increases in correlation. Additionally, CHIRPSv3-ERA5 and CHIRPSv3-IMERG show contrasting biases for CDD (underestimation) and CWD (overestimation). The underestimation of CDD (mBIAS < 0) by both CHIRPSv3 products is widespread across the country, whereas CHIRPSv2 shows underestimation concentrated in the Caribbean and Orinoco regions (Fig. S4.11). For CWD, CHIRPSv2, CHIRPSv3-ERA5, and CHIRPSv3-IMERG exhibit substantial overestimation (mBIAS $> 75\%$) in the Caribbean region, consistent with systematic overestimation in the number of light precipitation events (fBIAS > 3) across this region for the three products (Fig. S4.5). In the remaining natural regions, CHIRPSv3 variants and CHIRPSv2 show opposite patterns. While CHIRPSv2 tends to underestimate the CWD length, both CHIRPSv3 products, particularly CHIRPSv3-ERA5, overestimate CWD in the Amazon, Andes, Pacific, and Orinoco regions (Fig. S4.11). This pattern is again associated with systematic overestimation of the number of light precipitation events for CHIRPSv3 variants, whereas these events are underestimated in CHIRPSv2 across these regions (Figs. S4.1–S4.5). Finally, similar to CWD, all CHIRPS products exhibit comparable SDII patterns in the Caribbean region (mBIAS < 0), but show contrasting patterns elsewhere, with CHIRPSv2

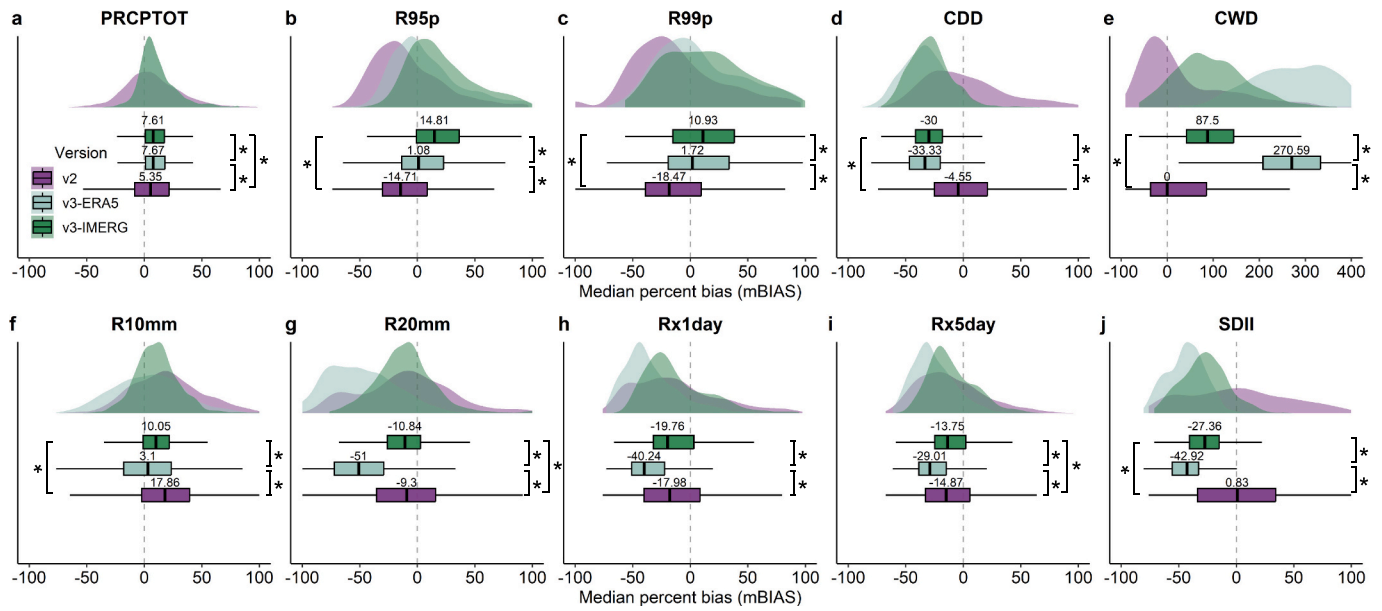


Fig. 4. Capability of CHIRPSv2 (v2), CHIRPSv3-ERA5 (v3-ERA5), and CHIRPSv3-IMERG (v3-IMERG) in capturing extreme precipitation indices based on the median percent bias (mBIAS). The grey dashed line indicates the optimal mBIAS value. Asterisks indicate statistically significant difference ($p < 0.05$) between CHIRPS products (e.g., CHIRPSv3-IMERG and CHIRPSv2 or CHIRPSv3-IMERG and CHIRPSv3-ERA5), based on a two-sided nonparametric Wilcoxon signed-rank test. PRCPTOT: total annual wet-day (≥ 1 mm) precipitation (mm), R95p: total annual precipitation from very wet days (mm), R99p: total annual precipitation from extreme wet days (mm), CDD: consecutive dry days (days), CWD: consecutive wet days (days), R10mm: number of days with precipitation ≥ 10 mm (days), R20mm: number of days with precipitation ≥ 20 mm (days), Rx1day: max 1-day precipitation amount (mm), Rx5day: max 5-day precipitation amount (mm), SDII: simple daily intensity index (mm/day). Further details in Table 2.

overestimating daily precipitation intensity and CHIRPSv3-ERA5 and CHIRPSv3-IMERG underestimating it.

3.3. Influence of ground stations density and wind-correction factor on CHIRPS performance

We examined how the performance of CHIRPSv2 and CHIRPSv3 at

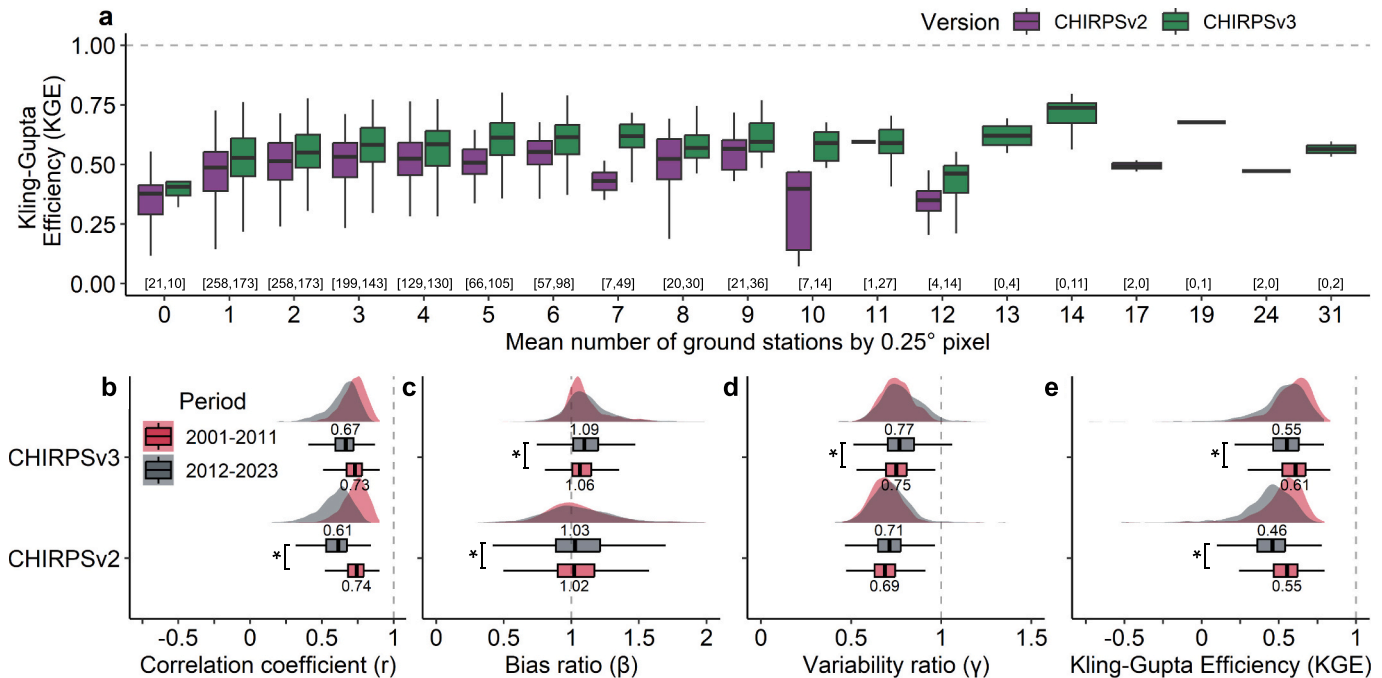


Fig. 5. (a) Pentad-scale performance of CHIRPSv2 and CHIRPSv3 as a function of the mean number of ground stations per 0.25° pixel used to generate each product. The grey dashed line indicates the optimal KGE value, and the values in brackets indicate the number of gauges used in our analysis for each number of stations category and CHIRPS version [v2, v3]. (b–e) Performance of CHIRPSv2 and CHIRPSv3 for the periods 2001–2011 and 2012–2023. The vertical grey line in panels b–e indicates the optimal value for each product and metric are shown above or below the corresponding boxplots. Asterisks indicate statistically significant difference ($p < 0.05$) between 2001 and 2011 and 2012–2023 periods for CHIRPSv2 and CHIRPSv3, based on a two-sided nonparametric Wilcoxon signed-rank test.

the pentad and monthly scales varies with the number of stations per 0.25° pixel used to produce each product (Figs. 5a and S5.1). The results show that the performance of CHIRPSv2 and CHIRPSv3 is lower in areas (0.25° pixels) without stations, compared to gauges located within pixels that contain at least one station at the pentad and monthly scales. These lower KGE values are mainly attributable to lower correlation coefficients, whereas bias and the variability ratio are less sensitive to changes in the number of ground stations (Fig. S5.2). Interestingly, our results suggest that both CHIRPSv2 and CHIRPSv3 exhibit similar performance when the number of stations ranges between 1 and 9. However, CHIRPSv3 overall outperforms CHIRPSv2 in areas with at least one ground station, demonstrating the effectiveness of the more accurate new CHPclimv2 and satellite-only CHIRPv3 products.

Due to the total number of available ground stations changing over time in both CHIRPS versions (Fig. S1.2), we also examined the performance of CHIRPSv2 and CHIRPSv3 separately for the periods 2001–2011 and 2012–2023 at pentad and monthly scales). Specifically, the mean number of ground stations decreases from 1935 to 746 (61%) in CHIRPSv2 and from 2516 to 1381 (45%) in CHIRPSv3 between these two periods. Our results indicate that both CHIRPSv2 and CHIRPSv3 exhibit better performance during the 2001–2011 period compared to the 2012–2023 period in terms of KGE values ($p < 0.05$) and correlation coefficients (Figs. 5b–5e). Specifically, CHIRPSv2 exhibits more pronounced decreases in performance between the two periods for both KGE and Pearson correlation. For both bias and variability ratios, the median differences between the two periods remain statistically significant, except for the CHIRPSv2 variability ratio, despite the relatively small magnitude of changes observed between 2001 and 2011 and 2012–2023 in both CHIRPSv2 and CHIRPSv3. Finally, we examined variations in CHIRPSv3 performance when IDEAM ground gauges were adjusted using the same monthly correction factors for each 0.25° pixel. Overall, the performance of CHIRPv3 shows small but statistically significant ($p < 0.05$) changes when ground gauges are wind-corrected or uncorrected at daily, pentad, or monthly scales across Colombia (Fig. 6). The most notable changes are observed in the bias ratio, which particularly decreases in the Caribbean region and mountain zones in the Andes region relative to the results from uncorrected gauge observations, resulting in a slight improvement in KGE values over these areas (Fig. S5.3). Finally, the categorical indices indicate that wind-bias correction does not substantially improve the ability of CHIRPSv3-IMERG or CHIRPSv3-ERA5 to capture daily precipitation events across different intensities (Fig. S5.4).

4. Discussion

4.1. Improvements in the CHIRP and CHIRPS products over Colombia

Our results indicate that the new CHIRPSv3 product more accurately captures precipitation magnitude and seasonality in Colombia compared to the CHIRPSv2, exhibiting an improved capability to capture high-range precipitation values measured by ground gauges at pentad, monthly, seasonal, and annual scales (Fig. 2 and Figs. S2.1–S2.2). For instance, CHPclimv2 provides a better estimation of observed monthly precipitation climatology across the country relative to CHPclimv1 ($KGE_{CHPclimv1} = 0.84$ and $KGE_{CHPclimv2} = 0.94$; Figs. S1.3–S1.4). This improvement is more evident in the Pacific region ($KGE_{CHPclimv1} = 0.33$ and $KGE_{CHPclimv2} = 0.86$), where CHPclimv1 largely underestimates ground observation estimates. This change in overall performance could be attributed to additional gauge-undercatch-corrected stations and data sources. For example, replacing TRMM-TMPA 3B42v7 with IMERGv6 in the new CHPclimv2 is expected to improve precipitation estimates due to its higher spatio-temporal resolution, which better captures precipitation in the Colombian Pacific Region (Palomino-Ángel et al., 2019). In this sense, the new CHIRPSv3 products also outperform CHIRPSv2 at the daily scale, particularly the variant that uses IMERG late version 7 data to disaggregate pentad CHIRPSv3 estimates into daily values (CHIRPSv3-IMERG; Figs. 3a–3d). This is consistent with the better performance of IMERG compared to ERA5 at the daily scale in Colombia (Valencia et al., 2023a). Additionally, the new satellite-only CHIRPv3 product exhibits higher linear correlation and lower RMSE compared to CHIRPv2 and CHIRPSv2, however, it also shows increased relative bias, with a consistent tendency toward overestimation across pentad to annual timescales (Figs. S2.1–S2.2).

CHIRPv3 and CHIRPSv3 show smaller underestimations of precipitation variability (Fig. 2 and Figs. S2.1–S2.2), which is one of the main limitations identified in CHIRPSv2 for Colombia (Baez-Villanueva et al., 2018; Valencia et al., 2023a) and many regions around the world (Alejo and Alejandro, 2021; Funk et al., 2020; Mulungu and Mukama, 2023; Paca et al., 2020). This is consistent with gauge validations in Ethiopia and Morocco, where CHIRPSv3 resolved the observed variability and 90th-percentile values better than CHIRPSv2 (Funk et al., 2026). This behavior is related to changes in the CHIRP algorithm. The intercept term of the CHIRPv2's regression-based CCD-to-precipitation translation partly suppressed the variance in v2. Additionally, the CHIRPv3 translation does not include an intercept term. Instead, a new slope is defined as the ratio of pentadal CHPclimv2 to the average CCD for that pentad. Additional constants are included to relax the slope value toward the

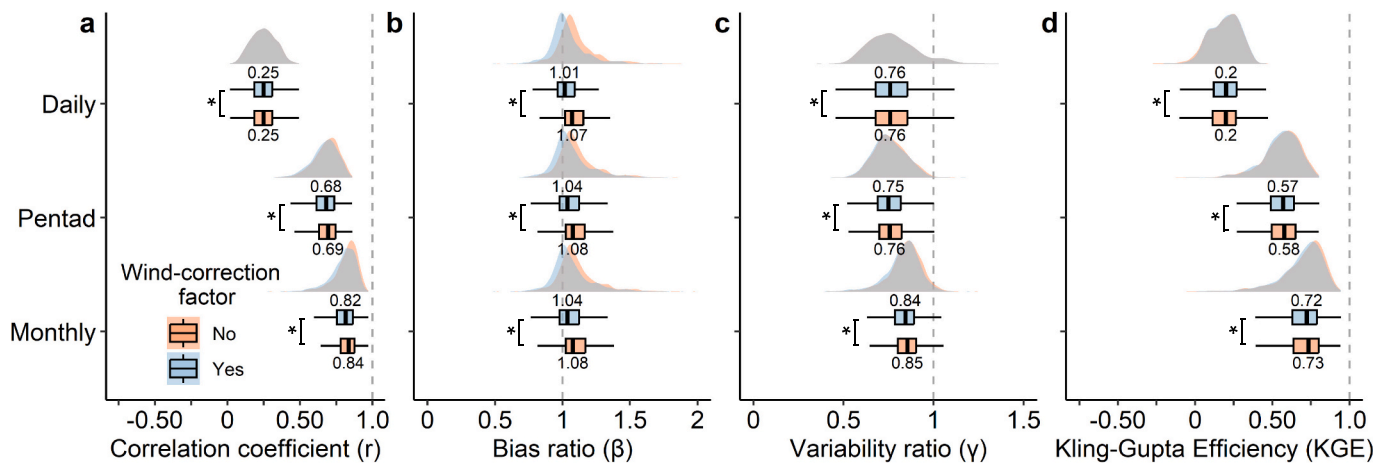


Fig. 6. Performance of CHIRPSv3 considering undercatch-corrected and uncorrected ground gauge observations at daily, pentad, and monthly scales. Daily CHIRPSv3 data were extracted from the CHIRPSv3-IMERG variant. Median values for each product and metric are shown above or below the corresponding boxplots. The vertical grey line in all panels indicates the optimal value for each index.

Geostationary Operational Environmental Satellite (GOES) precipitation index rate of 3 mm per hour as the climatology approaches zero (Funk et al., 2026).

Our results show that the performance of CHIRPSv2 and CHIRPSv3 at pentad and monthly scales is lower in areas (0.25° pixels) without any ground stations used in their estimation process, compared to locations where at least one station is located within the corresponding 0.25° pixel (Fig. 6b and Fig. S5.1). Particularly, the lack of stations in these pixels tends to reduce the correlation coefficients for both CHIRPSv2 and CHIRPSv3 (Fig. S5.2). In contrast, when the number of ground stations ranges from 1 to 9, there are no substantial differences in the performance of CHIRPSv2 (Fig. 6a) and CHIRPSv3 (Fig. 5b). This could be attributed to the modified inverse distance-weighted averaging process used to blend station observations with CHIRP estimates in the generation of CHIRPS (Funk et al., 2026). As the distance from pixel i to ground station j increases, the influence of that station on the CHIRP bias correction may diminish, potentially reducing the CHIRP accuracy estimates. Moreover, since the distance-weighted average is based only on the five nearest stations to pixel i , increasing station density beyond this threshold does not necessarily lead to a substantial improvement in the final CHIRPS estimates. Indeed, our results suggest that having 3–5 stations per pixel leads to a median KGE above 0.5 for both CHIRPSv2 and CHIRPSv3 (Figs. 5a–5b). The influence of the number of ground stations used in the development of CHIRPSv2 and CHIRPSv3 is also observed by comparing the performance of each product for the periods 2001–2011 and 2012–2023. Overall, our results show that both products exhibit a lower performance at pentad and monthly scales over the 2012–2023 period (Fig. 5 and Fig. S5.1), in which the number of ground stations had a substantial decrease (61% in v2 and 45% in v3) compared with the period 2001–2011 across the country (Fig. S1.2a). This highlights the importance of ground stations with long-term observations in improving the performance of multi-source precipitation products such as CHIRPS. In this sense, although the number of precipitation gauges has increased in Colombia and in many regions around the world, the number of stations incorporated into CHIRPS has declined significantly in recent decades. Thus, further efforts by national weather services are needed to facilitate the integration of ground observations into global precipitation products.

Regarding topographic and climatic gradients, our findings demonstrate that CHIRPv3 and CHIRPSv3 capture more accurately the spatio-temporal variability of precipitation across elevation ranges and MAP levels in Colombia (Figs. S3.2–3.3 and S4.6–4.8). While the performance of CHIRPv2 and CHIRPSv2 exhibits an elevation-bias dependence, with greater uncertainty in mountain areas (Mulungu and Mukama, 2023; Peña-Guerrero et al., 2022), CHIRPv3 and CHIRPSv3 show more consistent results from low to high elevations at daily, pentad, and monthly scales. This improvement may be attributed to the incorporation of IMERG inputs at 0.10° resolution, rather than TRMM at 0.25° resolution, in CHIRP estimates. Indeed, the accuracy of remote sensing precipitation products in mountainous areas is largely influenced by spatial resolution (Baez-Villanueva et al., 2018). Also, the inclusion of the wind-bias correction may help to reduce this bias, especially in regions such as the Andes and Caribbean that show the highest monthly wind-correction factor values (Fig. S1.5). In this sense, our results show slight improvements in terms of the bias ratio at high elevations when ground gauges are wind-bias corrected (Fig. S5.3). In general, CHIRPv2 and CHIRPSv2 tend to underestimate precipitation magnitude for regions with MAP above 3000 mm (Paca et al., 2020; and Figs. S2.1 and S3.1–3.2). Conversely, CHIRPv3 and CHIRPSv3 exhibit a lower dependence on precipitation gradients (Fig. S3.2), which could be explained by improvements in CHPclimv2 and CHIRPv3 component estimates, particularly in the Pacific region — the wettest part of the country and one of the wettest areas in the world (Poveda et al., 2025).

Ultimately, although our analysis focuses on comparing CHIRPSv3 with CHIRPSv2, CHIRPSv3 also tends to outperform MSWEPv2.8 and IMERG Final Run across the country, except in the Amazon region, based

on the results of our previous work (Valencia et al., 2023a).

4.2. Persistent limitations in CHIRPSv3

The main persistent limitations in CHIRPSv3 are observed at the daily scale (Fig. 3a–3d), in which the improvements relative to CHIRPSv2 are minimal, exhibiting low correlation (Median $r_{\text{CHIRPSv2}} = 0.18$ and Median $r_{\text{CHIRPSv3}} = 0.25$ to 0.30) and KGE values (Median $\text{KGE}_{\text{CHIRPSv2}} = 0.12$ and Median $\text{KGE}_{\text{CHIRPSv3}} = 0.17$ to 0.25) (Figs. 3a–3d). Our results show similar patterns in the performance of CHIRPSv3 in capturing different daily precipitation intensities and extreme precipitation indices (Figs. 3e–3h and 4). These limitations may be attributed to the CHIRP and CHIRPS data sources, their algorithms, and shortcomings in ground-based observations. For instance, a temporal mismatch between CHIRPS and ground gauges introduces bias in our analysis. Specifically, daily gauge measurements in Colombia represent 24-h accumulations from 12:00 UTC on day i to 12:00 UTC on day $i + 1$, whereas CHIRPS daily estimates correspond to the 00:00 to 23:59 UTC period. Thus, the performance of gridded precipitation products improves significantly when daily observations are aggregated to a 3-day scale (Gómez et al., 2023; Valencia et al., 2023a) or to the pentad scale (Fig. 2). Moreover, although most of the evaluated gauges have a low percentage of missing values (Fig. S1.1), these data gaps still may influence the performance of CHIRPS, especially regarding extreme precipitation indices.

CHIRPSv2, like other gridded precipitation products, has a relatively good capability in capturing non-precipitation events (< 1 mm/day), but limited representation of light, moderate, heavy, and violent precipitation events in Colombia and globally (Baez-Villanueva et al., 2018; Mekonnen et al., 2023; Valencia et al., 2023a). Particularly, both CHIRPSv2 and CHIRPSv3, by construction, are expected to be less able to accurately capture daily precipitation events, as these products are primarily generated at the pentad scale and then partitioned into daily time steps using CCD. The new CHIRPSv3 product continues to exhibit these limitations and, in some cases, further exacerbates them, showing larger overestimations or underestimations in the number of precipitation events compared to CHIRPSv2 (Fig. 3). For instance, both CHIRPSv3-ERA5 and CHIRPSv3-IMERG exhibit larger underestimations ($\text{fBIAS} < 1$) and overestimations ($\text{fBIAS} > 1$) in the number of non-precipitation and light precipitation events, respectively; across the country (Fig. 3). Notably, our results show the largest overestimations in the number of light precipitation events during the dry season and across the Caribbean region. Similarly, the results of extreme precipitation indices show that CHIRPSv3, like CHIRPSv2, underestimates the frequency (R20mm) and intensity (Rx1day , Rx5day , and SDII) of extreme precipitation events (Fig. 4); as previously reported for CHIRPSv2 in Brazil (Cavalcante et al., 2020) and Nicaragua (Mendez-Rivas et al., 2024). Dubache et al. (2026) also report low capability of both CHIRPSv3-ERA5 and CHIRPSv3-IMERG ($\text{POD} < 0.45$ and $\text{FAR} > 0.60$) to represent daily precipitation events exceeding 10 mm/day in Benin, West Africa. These limitations may reduce the suitability of CHIRPSv3 for applications requiring an accurate representation of high-intensity precipitation events, such as high floods and soil erosion modeling, and the development of intensity–duration–frequency relationships (Alarcón Hurtado et al., 2026). Furthermore, the substantial decline in CHIRPSv3 performance in capturing CWD and CCD, relative to CHIRPSv2, may also limit its applicability to early-warning systems (e.g., landslides and wildfires).

Three main factors in the CHIRPS formulation may contribute to the overestimation and underestimation of dry days and precipitation events. First, the gauge-undercatch correction applied in the estimation of the new CHPclimv2 and CHIRPv3 may inflate the occurrence of light precipitation, as evidenced by the larger overestimations in the number of such events when comparing uncorrected and wind-corrected ground gauge observations (Fig. S5.4). Second, daily IMERG and ERA5 also exhibit limitations in reliably capturing non-precipitation and light

precipitation events in Colombia (Baez-Villanueva et al., 2018; Valencia et al., 2023a), potentially introducing similar issues into CHIRPSv3-ERA5 and CHIRPSv3-IMERG when disaggregating pentad estimates to daily resolution. Third, although CHIRPv3 uses a new formulation that captures a wider range of precipitation values, its estimates largely depend on cold cloud duration observations (cloud-top brightness < 235 K) from geostationary satellite thermal infrared (TIR) observations (Funk et al., 2026). The TIR method is based on the premise that cloud-top temperature is an indicator of cloud-top height, and that higher, colder clouds result in more precipitation. However, this is an indirect relationship, as colder clouds do not always produce precipitation (So and Shin, 2018; Varma, 2018). For example, in-situ cirrus or remnants of anvil cirrus may form without underlying precipitation, collision-coalescence in warm-phase clouds can still cause intense rainfall, or the coldest cloud-top temperatures may be spatially offset from underlying precipitation due to wind shear. The assumed link between convective intensity and precipitation intensity may contribute to the weaker performance of CHIRP and CHIRPS products on extreme precipitation metrics (Hamada et al., 2015).

These limitations of the TIR method in estimating precipitation may be more pronounced in areas with frequent and large mesoscale convective systems (MCSs), such as the Caribbean region (Hernandez-Deckers, 2022). Indeed, both CHIRPSv2 and CHIRPSv3 show systematic underestimation of non-precipitation events and an overestimation of light and moderate precipitation events in this region (Fig. S4.11). Although TIR observations are commonly used to capture the lifecycle evolution of MCSs using cloud-top brightness temperature, this approach lacks information on their microphysical structure, including size and phase of hydrometeors at different depths within the storms (Feng et al., 2021). Underestimates of occurrence or overestimates of precipitation intensity from MCSs could explain the relatively poorer performance of CHIRPSv3 in the Pacific and Caribbean regions, where these systems can occur up to 20% of the time during boreal summer (Gilmour et al., 2026). For instance, CHIRPSv3 exhibits the largest overestimations of precipitation magnitude in the Caribbean region at both pentad and monthly scales (Figs. S2.3 and 2.6). Additionally, while CHIRPSv3 is the best product ($KGE_{CHIRPSv3} > KGE_{CHIRPSv2}$) at more than 80% of the evaluated gauges across each natural region in Colombia, it is the best product at only ~66% of the gauges in the Caribbean region. Additionally, the strong, localized topography bordering the Pacific region poses additional challenges for accurately representing MSCs. In contrast, gauge scarcity likely drives poorer performance across the Amazon region (Fig. 1d), while the dominance of warm rain (CCD > 273 K) further limits the TIR algorithm. Further analyses are required to examine how MCSs influence CHIRPSv3 estimates, as precipitation associated with MCSs represents a significant fraction of total seasonal and annual precipitation across the tropics (Feng et al., 2021).

4.3. Study limitations and future directions

Our analysis included both independent and non-independent gauges used in CHIRPS development, which may inflate performance metrics, especially when compared to satellite and reanalysis products (Vallejo-Bernal et al., 2021). We selected both categories of gauges for three main reasons: (i) the number of gauges blended with CHIRP estimates to produce CHIRPS varies monthly due to data availability and quality control, (ii) non-independent gauges provide an opportunity to verify the improvement of CHIRP bias correction and its dependence on station density, (iii) we compared the performance of CHIRPSv3 relative to CHIRPSv2, both of which blending several ground gauges (Fig. S1.2). Overall, our results consistently show that CHIRPSv3 outperforms CHIRPSv2 across independent pixels and in areas with 1 to 9 blended stations (Figs. 4 and S5.1–5.2). Further analyses are needed to comprehensively assess the performance of the new CHIRPSv3 product in comparison with other gridded precipitation products, such as IMERG Final Run and MSWEPv2.8 (including its recent update, MSWEPv3;

Wang et al., 2026), which have previously been identified as among the best-performing products in Colombia (Baez-Villanueva et al., 2018; Valencia et al., 2023a) and globally (Beck et al., 2019; Huffman et al., 2020; Vargas Godoy et al., 2025).

The point-to-pixel analysis is commonly used to evaluate the performance of gridded precipitation products (e.g., Baez-Villanueva et al., 2018; Mekonnen et al., 2023; Valencia et al., 2023a; Zambrano-Bigiarini et al., 2017). However, this approach may introduce bias in our analysis due to the mismatches in spatial scale. For instance, ground gauges provide precipitation measurements at a single location or point, while pixels represent the average precipitation over a larger area (~30 km² in the CHIRP and CHIRPS products). Thus, this mismatch can substantially affect performance, particularly in regions with low gauge density or high spatial precipitation variability, such as mountainous areas. Although pixel-to-pixel analysis is an alternative to the point-to-pixel approach, it tends to greater dependence on gauge distribution because it requires interpolation to match ground observations to satellite pixels for consistent comparisons.

Finally, the new CHIRPSv3 product has the potential to improve climatic, ecological, and hydrological studies across the tropics compared to the v2, as indicated by our results and performance evaluations in Central America, Ecuador, and Brazil (Funk et al., 2026). However, its poor performance at the daily scale (Figs. 3 and 4) may introduce biases in applications requiring daily precipitation estimates, including hydrological modeling and landslide forecasting. This is attributed to the fact that CHIRPS is primarily generated at the pentad scale and then partitioned into daily time steps using CCD. Conversely, CHIRPSv3 provides high-quality precipitation estimates at the finest spatial resolution (0.05°), compared to MSWEP and IMERG (0.10°), for studies spanning pentad to annual temporal scales, such as drought monitoring. Further evaluation of CHIRPSv3 for hydrological modeling, landslide forecasting, and drought monitoring applications represents a valuable direction for future research.

5. Conclusions

This study evaluates the performance of the new CHIRPSv3 products relative to CHIRPSv2, examining the improvements and persistent limitations across multiple temporal and spatial scales for Colombia — a country characterized by complex topographic and climatic gradients representative of those found across the tropics. Our results indicate that CHIRPSv3 outperforms CHIRPSv2 across daily to annual scales and at both regional and national levels, exhibiting significantly higher median Kling-Gupta Efficiency (KGE) values. Similarly, the satellite-only CHIRPv3 product exhibits better performance than CHIRPv2. In particular, version 3 products exhibit higher temporal correlation and lower underestimation of precipitation variability, providing a more accurate representation of higher precipitation values, including those in the wettest regions of the country. Additionally, the new versions of CHIRP and CHIRPS (v3) perform better and exhibit lower sensitivity to topographic gradients than their previous versions (v2), which show an elevation-dependent bias. These improvements result from a more accurate climatology component (CHPclimv2) that incorporates a greater number of wind-bias-corrected ground-based observations in the new CHPclimv2, CHIRPv3, and CHIRPSv3 formulations. Moreover, the updated algorithm for estimating the satellite-only CHIRPv3 product more effectively captures a wider range of precipitation values. However, despite improvements in CHIRPSv3 relative to CHIRPSv2, a low capability persists in capturing daily precipitation, particularly in identifying both dry days (< 1 mm/day) and precipitation events (≥ 1 mm/day) in Colombia. Our results also indicate that CHIRPSv3 still shows limited accuracy in capturing extreme indices used to characterize the duration (CDD, CWD) and intensity (Rx1day and SDII) of precipitation. These results demonstrated the overall effectiveness of the new CHIRPSv3 formulation, improving the representation of spatio-temporal precipitation variability, a critical factor for climatic,

ecological, and hydrological studies.

CRediT authorship contribution statement

Santiago Valencia: Formal analysis, Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Diver E. Marín:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Derly Gómez:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Valentina Echavarría-Porras:** Writing – review & editing, Methodology. **Juliana Mejía-Sepúlveda:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Admin Husic:** Writing – review & editing, Investigation. **Sylvia Sullivan:** Writing – review & editing, Methodology, Investigation. **Natalia Hoyos:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Juan Camilo Villegas:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Laura Harrison:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Julia K. Green:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atmosres.2026.108971>.

Data availability

The main R code and datasets required to reproduce the analysis, as well as the main figures, are available via Zenodo at <https://doi.org/10.5281/zenodo.19150519>. The complete IDEAM dataset is available at <http://dhime.ideam.gov.co/atencionciudadano/>. The CHIRP and CHIRPS data, as well as wind-bias correction factors, can be obtained from <https://data.chc.ucs.edu/products/>.

References

- Ahmed, J.S., Buizza, R., Dell'Acqua, M., Demissie, T., Pè, M.E., 2024. Evaluation of ERA5 and CHIRPS rainfall estimates against observations across Ethiopia. *Meteorol. Atmos. Phys.* 136, 17. <https://doi.org/10.1007/s00703-024-01008-0>.
- Aksu, H., Akgül, M.A., 2020. Performance evaluation of CHIRPS satellite precipitation estimates over Turkey. *Theor. Appl. Climatol.* 142, 71–84. <https://doi.org/10.1007/s00704-020-03301-5>.
- Alarcón Hurtado, E.A., Yanala Bravo, J.C., Jimenez Osorio, D.A., Avila-Diaz, A., 2026. Performance of high-resolution gridded precipitation products for determining long period return values and standardized precipitation index in Colombia. *Hydrol. Sci. J.* 71, 602–619. <https://doi.org/10.1080/02626667.2025.2600094>.
- Alejo, L.A., Alejandro, A.S., 2021. Validating CHIRPS ability to estimate rainfall amount and detect rainfall occurrences in the Philippines. *Theor. Appl. Climatol.* 145, 967–977. <https://doi.org/10.1007/s00704-021-03685-y>.
- Arias, P.A., Rivera, J.A., Sörensson, A.A., Zachariah, M., Barnes, C., Philip, S., Kew, S., Vautard, R., Koren, G., Pinto, I., Vahlberg, M., Singh, R., Raju, E., Li, S., Yang, W., Vecchi, G.A., Otto, F.E.L., 2024. Interplay between climate change and climate variability: the 2022 drought in central South America. *Clim. Chang.* 177. <https://doi.org/10.1007/s10584-023-03664-4>.
- Ayoub, A.B., Tangang, F., Juneng, L., Tan, M.L., Chung, J.X., 2020. Evaluation of gridded precipitation datasets in Malaysia. *Remote Sens.* 12, 613. <https://doi.org/10.3390/rs12040613>.
- Baez-Villanueva, O.M., Zambrano-Bigiarini, M., Ribbe, L., Nauditt, A., Giraldo-Osorio, J.D., Thinh, N.X., 2018. Temporal and spatial evaluation of satellite rainfall estimates over different regions in Latin-America. *Atmos. Res.* 213, 34–50. <https://doi.org/10.1016/j.atmosres.2018.05.011>.
- Beck, H.E., Wood, E.F., Pan, M., Fisher, C.K., Miralles, D.G., Van Dijk, A.I.J.M., McVicar, T.R., Adler, R.F., 2019. MSWEP v2 global 3-Hourly 0.1° precipitation: Methodology and quantitative assessment. *Bull. Am. Meteorol. Soc.* 100, 473–500. <https://doi.org/10.1175/BAMS-D-17-0138.1>.
- Cavalcante, R.B.L., Ferreira, D.B.D.S., Pontes, P.R.M., Tedeschi, R.G., Da Costa, C.P.W., De Souza, E.B., 2020. Evaluation of extreme rainfall indices from CHIRPS precipitation estimates over the Brazilian Amazonia. *Atmos. Res.* 238, 104879. <https://doi.org/10.1016/j.atmosres.2020.104879>.
- Cepeda Arias, E., Cañon Barriga, J., 2022. Performance of high-resolution precipitation datasets CHIRPS and TerraClimate in a Colombian high Andean Basin. *Geocarto Int.* 37, 17382–17402. <https://doi.org/10.1080/10106049.2022.2129816>.
- Dinku, T., Funk, C., Peterson, P., Maidment, R., Tadesse, T., Gadain, H., Ceccato, P., 2018. Validation of the CHIRPS satellite rainfall estimates over eastern Africa. *Quart. J. Royal Meteorol. Soc.* 144, 292–312. <https://doi.org/10.1002/qj.3244>.
- Du, H., Tan, M.L., Zhang, F., Chun, K.P., Li, L., Kabir, M.H., 2024. Evaluating the effectiveness of CHIRPS data for hydroclimatic studies. *Theor. Appl. Climatol.* 155, 1519–1539. <https://doi.org/10.1007/s00704-023-04721-9>.
- Dubache, G., Kakpa, D., Measho Yhdego, S., Li, Z., Ntwali, D., Hirwa, H., Bekalo Sappa, L., 2026. Reliability of modern satellite precipitation products in a data-scarce West African country: A comprehensive validation for Benin. *Quart. J. Royal Meteorol. Soc.* 70120. <https://doi.org/10.1002/qj.70120>.
- Escobar, M., Hoyos, I., Nieto, R., Villegas, J.C., 2022. The importance of continental evaporation for precipitation in Colombia: a baseline combining observations from stable isotopes and modelling moisture trajectories. *Hydrol. Process.* 36, e14595. <https://doi.org/10.1002/hyp.14595>.
- Espinoza, J.C., Garreaud, R., Poveda, G., Arias, P.A., Molina-Carpio, J., Masiokas, M., Viale, M., Scaff, L., 2020. Hydroclimate of the Andes part I: main climatic features. *Front. Earth Sci.* 8, 64. <https://doi.org/10.3389/feart.2020.00064>.
- Feldman, A.F., Feng, X., Felton, A.J., Crow, W.T., Biederman, J.A., Konings, A.G., Poulter, B., Serbin, S.P., 2026. Widespread Co-location of less frequent and more intense daily precipitation over land. *Geophys. Res. Lett.* 53, e2025GL120745. <https://doi.org/10.1029/2025GL120745>.
- Feng, Z., Leung, L.R., Liu, N., Wang, J., Houze, R.A., Li, J., Hardin, J.C., Chen, D., Guo, J., 2021. A global high-resolution mesoscale convective system database using satellite-derived cloud tops, surface precipitation, and tracking. *JGR-Atmos.* 126, e2020JD034202. <https://doi.org/10.1029/2020JD034202>.
- Fick, S.E., Hijmans, R.J., 2017. WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. *Int. J. Climatol.* 37, 4302–4315. <https://doi.org/10.1002/joc.5086>.
- Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland, J., Harrison, L., Hoell, A., Michaelsen, J., 2015. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. *Sci. Data* 2, 150066. <https://doi.org/10.1038/sdata.2015.66>.
- Funk, C., Peterson, P., Landsfeld, M., Davenport, F., Becker, A., Schneider, U., Pedreros, D., McNally, A., Arsenault, K., Harrison, L., Shukla, S., 2020. Algorithm and data improvements for version 2.1 of the climate hazards center's infrared precipitation with stations data set. In: Levizzani, V., Kidd, C., Kirschbaum, D.B., Kummerow, C.D., Nakamura, K., Turk, F.J. (Eds.), *Satellite Precipitation Measurement, Advances in Global Change Research*. Springer International Publishing, Cham, pp. 409–427. https://doi.org/10.1007/978-3-030-24568-9_23.
- Funk, C., Peterson, P., Harrison, L., Saldivar, R., Landsfeld, M., Pedreros, D., Shukla, S., Fink, A.H., Davenport, F., Peterson, S., Turner, W., Sonnier, A., Budde, M., Tabor, K., Verdin, J., Hauzaree, D., Naim, M., Alaso, D., Husak, G., 2026. The climate hazards center infrared precipitation with stations, version 3. In: *Sci. Data*. [10.1038/s41597-026-07096-4](https://doi.org/10.1038/s41597-026-07096-4).
- Funk, C., Peterson, P., Huffman, G.J., Landsfeld, M.F., Peters-Lidard, C., Davenport, F., Shukla, S., Peterson, S., Pedreros, D.H., Ruane, A.C., Muttter, C., Turner, W., Harrison, L., Sonnier, A., Way-Henthorne, J., Husak, G.J., 2022. Introducing and evaluating the climate hazards center IMERG with stations (CHIMES): Timely Station-Enhanced integrated multisatellite retrievals for global precipitation measurement. *Bull. Am. Meteorol. Soc.* 103, E429–E454. <https://doi.org/10.1175/bams-d-20-0245.1>.
- Gebrechorkos, S., Leyland, J., Slater, L., Wortmann, M., Ashworth, P.J., Bennett, G.L., Boothroyd, R., Cloke, H., Delorme, P., Griffith, H., Hardy, R., Hawker, L., McLelland, S., Neal, J., Nicholas, A., Tatem, A.J., Vahidi, E., Parsons, D.R., Darby, S.E., 2023. A high-resolution daily global dataset of statistically downscaled CMIP6 models for climate impact analyses. *Sci. Data* 10, 611. <https://doi.org/10.1038/s41597-023-02528-x>.
- Gilmour, H., Chadwick, R., Catto, J.L., Halladay, K., Hart, N.C.G., Rehbein, A., 2026. Mesoscale convective systems over south america: Representation in Kilometer-Scale met office unified model climate simulations. *J. Clim.* 39, 225–247. <https://doi.org/10.1175/JCLI-D-24-0754.1>.
- Gómez, D., Aristizábal, E., García, E.F., Marín, D., Valencia, S., Vásquez, M., 2023. Landslides forecasting using satellite rainfall estimations and machine learning in the colombian andean region. *J. S. Am. Earth Sci.* 125, 104293. <https://doi.org/10.1016/j.jsames.2023.104293>.
- Gupta, H.V., Kling, H., Yilmaz, K.K., Martinez, G.F., 2009. Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *J. Hydrol.* 377, 80–91. <https://doi.org/10.1016/j.jhydrol.2009.08.003>.
- Hamada, A., Takayabu, Y.N., Liu, C., Zipser, E.J., 2015. Weak linkage between the heaviest rainfall and tallest storms. *Nat. Commun.* 6, 6213. <https://doi.org/10.1038/ncomms7213>.

- Harrison, L., Funk, C., Peterson, P., 2019. Identifying changing precipitation extremes in Sub-Saharan Africa with gauge and satellite products. *Environ. Res. Lett.* 14, 085007. <https://doi.org/10.1088/1748-9326/ab2cae>.
- Hernandez-Deckers, D., 2022. Features of atmospheric deep convection in northwestern South America obtained from infrared satellite data. *Quart J Royal Meteorol Soc* 148, 338–350. <https://doi.org/10.1002/qj.4208>.
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R.J., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., De Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., Thépaut, J., 2020. The ERA5 global reanalysis. *Quart J Royal Meteorol Soc* 146, 1999–2049. <https://doi.org/10.1002/qj.3803>.
- Huffman, G.J., Bolvin, D.T., Braithwaite, D., Hsu, K.-L., Joyce, R.J., Kidd, C., Nelkin, E.J., Sorooshian, S., Stocker, E.F., Tan, J., Wolff, D.B., Xie, P., 2020. Integrated Multi-satellite Retrievals for the Global Precipitation Measurement (GPM) Mission (IMERG). In: Levizzani, V., Kidd, C., Kirschbaum, D.B., Kummerow, C.D., Nakamura, K., Turk, F.J. (Eds.), *Satellite Precipitation Measurement, Advances in Global Change Research*. Springer International Publishing, Cham, pp. 343–353. https://doi.org/10.1007/978-3-030-24568-9_19.
- IGAC, 1997. Mapa de regiones naturales de Colombia. Escala 1: 5.000. 000. SIGAC. Instituto Geográfico Agustín Codazzi, Bogotá.
- Jarvis, A., Reuter, H.I., Nelson, A., Guevara, E., 2008. Hole-filled SRTM for the globe version 4. In: Available from the CGIAR-CSI SRTM 90m Database (<http://srtm.csi.cgiar.org>) 15, 5.
- Kling, H., Fuchs, M., Paulin, M., 2012. Runoff conditions in the upper Danube basin under an ensemble of climate change scenarios. *J. Hydrol.* 424–425, 264–277. <https://doi.org/10.1016/j.jhydrol.2012.01.011>.
- Legates, D.R., Willmott, C.J., 1990. Mean seasonal and spatial variability in gauge-corrected, global precipitation. *Int. J. Climatol.* 10, 111–127. <https://doi.org/10.1002/joc.3370100202>.
- Lehner, B., Grill, G., 2013. Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrol. Process.* 27, 2171–2186. <https://doi.org/10.1002/hyp.9740>.
- López-Bermeo, C., Montoya, R.D., Caro-Lopera, F.J., Díaz-García, J.A., 2022. Validation of the accuracy of the CHIRPS precipitation dataset at representing climate variability in a tropical mountainous region of South America. *Phys. Chem. Earth, Parts A/B/C* 127, 103184. <https://doi.org/10.1016/j.pce.2022.103184>.
- Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S.L., Péan, C., Berger, S., Caud, N., Chen, Y., Goldfarb, L., Gomis, M., 2021. *Climate change 2021: the physical science basis. In: Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change 2*, 2391.
- Mekonnen, K., Velpuri, N.M., Leh, M., Akpoti, K., Owusu, A., Tinonetsana, P., Hamouda, T., Ghansah, B., Paranamana, T.P., Munzimi, Y., 2023. Accuracy of satellite and reanalysis rainfall estimates over Africa: a multi-scale assessment of eight products for continental applications. *J. Hydrol. Reg. Stud.* 49, 101514. <https://doi.org/10.1016/j.ejrh.2023.101514>.
- Mendez-Rivas, R., Palacios, M.M., Lemus, R.P., 2024. Spatiotemporal variability of extreme rainfall indices over Nicaragua between 2001 and 2016 and their relationship with teleconnection patterns. *J. Appl. Meteorol. Climatol.* 63, 321–338. <https://doi.org/10.1175/JAMC-D-22-0112.1>.
- Mulungu, D.M., Mukama, E., 2023. Evaluation and modelling of accuracy of satellite-based CHIRPS rainfall data in Ruwusu subbasin, Tanzania. *Model. Earth Syst. Environ.* 9, 1287–1300. <https://doi.org/10.1007/s40808-022-01555-8>.
- Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D.G., Piles, M., Rodríguez-Fernández, N.J., Zsoter, E., Buontempo, C., Thépaut, J.-N., 2021. ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth Syst. Sci. Data* 13, 4349–4383. <https://doi.org/10.5194/essd-13-4349-2021>.
- Nguyen, P., Ombadi, M., Sorooshian, S., Hsu, K., AghaKouchak, A., Braithwaite, D., Ashouri, H., Thorstensen, A.R., 2018. The PERSIANN family of global satellite precipitation data: a review and evaluation of products. *Hydrol. Earth Syst. Sci.* 22, 5801–5816. <https://doi.org/10.5194/hess-22-5801-2018>.
- Nicholson, S.E., Klotter, D.A., 2021. Assessing the reliability of satellite and reanalysis estimates of rainfall in equatorial Africa. *Remote Sens.* 13, 3609. <https://doi.org/10.3390/rs13183609>.
- Ocampo-Marulanda, C., Fernández-Álvarez, C., Cerón, W.L., Canchala, T., Carvajal-Escobar, Y., Alfonso-Morales, W., 2022. A spatiotemporal assessment of the high-resolution CHIRPS rainfall dataset in southwestern Colombia using combined principal component analysis. *Ain Shams Eng. J.* 13, 101739. <https://doi.org/10.1016/j.asej.2022.101739>.
- Paca, V.H.D.M., Espinoza-Dávalos, G., Moreira, D., Comair, G., 2020. Variability of trends in precipitation across the Amazon river basin determined from the CHIRPS precipitation product and from station records. *Water* 12, 1244. <https://doi.org/10.3390/w12051244>.
- Palomino-Ángel, S., Anaya-Acevedo, J.A., Botero, B.A., 2019. Evaluation of 3B42V7 and IMERG daily-precipitation products for a very high-precipitation region in northwestern South America. *Atmos. Res.* 217, 37–48. <https://doi.org/10.1016/j.atmosres.2018.10.012>.
- Paredes-Trejo, F.J., Barbosa, H.A., Lakshmi Kumar, T.V., 2017. Validating CHIRPS-based satellite precipitation estimates in Northeast Brazil. *J. Arid Environ.* 139, 26–40. <https://doi.org/10.1016/j.jaridenv.2016.12.009>.
- Peña-Guerrero, M.D., Umirbekov, A., Tarasova, L., Müller, D., 2022. Comparing the performance of high-resolution global precipitation products across topographic and climatic gradients of Central Asia. *Int. J. Climatol.* 42, 5554–5569. <https://doi.org/10.1002/joc.7548>.
- Peng, J., Dadson, S., Hirpa, F., Dyer, E., Lees, T., Miralles, D.G., Vicente-Serrano, S.M., Funk, C., 2020. A pan-African high-resolution drought index dataset. *Earth Syst. Sci. Data* 12, 753–769. <https://doi.org/10.5194/essd-12-753-2020>.
- Poveda, G., Mejía, J.F., Arias, P.A., Zuluaga, M.D., Salazar, J.F., Carmona, A.M., Builes-Jaramillo, A., Salas, H.D., Yepes, J., Martínez, J.A., Bedoya-Soto, J.M., 2025. Climate of Northwestern South America. In: *Oxford Research Encyclopedia of Climate Science*. Oxford University Press. <https://doi.org/10.1093/acrefore/9780190228620.013.966>.
- Pradhan, R.K., Markonis, Y., Vargas Godoy, M.R., Villalba-Pradas, A., Andreadis, K.M., Nikolopoulos, E.I., Papalexiou, S.M., Rahim, A., Tapiador, F.J., Hanel, M., 2022. Review of GPM IMERG performance: a global perspective. *Remote Sens. Environ.* 268, 112754. <https://doi.org/10.1016/j.rse.2021.112754>.
- R Core Team, 2023. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <http://www.R-project.org/>.
- Ribeiro, A.F.S., Santos, L., Randerson, J.T., Uribe, M.R., Alencar, A.A.C., Macedo, M.N., Morton, D.C., Zscheischler, J., Silvestrini, R.A., Rattis, L., Seneviratne, S.I., Brando, P.M., 2024. The time since land-use transition drives changes in fire activity in the Amazon-Cerrado region. *Commun. Earth Environ.* 5. <https://doi.org/10.1038/s43247-024-01248-3>.
- Rivera, J.A., Marianetti, G., Hinrichs, S., 2018. Validation of CHIRPS precipitation dataset along the Central Andes of Argentina. *Atmos. Res.* 213, 437–449. <https://doi.org/10.1016/j.atmosres.2018.06.023>.
- Romero-Hernández, C.M., Avila-Díaz, A., Quesada, B., Medeiros, F., Cerón, W.L., Guzmán-Escalante, J., Ocampo-Marulanda, C., Rodríguez Torres, R., Zuluaga, C.F., 2024. Bias-corrected high-resolution precipitation datasets assessment over a tropical mountainous region in Colombia: a case of study in upper Cauca river basin. *J. S. Am. Earth Sci.* 140, 104898. <https://doi.org/10.1016/j.jsames.2024.104898>.
- Sa'adi, Z., Yusop, Z., Alias, N.E., Shiru, M.S., Muhammad, M.K.I., Ramli, M.W.A., 2023. Application of CHIRPS dataset in the selection of rain-based indices for drought assessments in Johor river basin, Malaysia. *Sci. Total Environ.* 892, 164471. <https://doi.org/10.1016/j.scitotenv.2023.164471>.
- Saeidizand, R., Sabetghadam, S., Tarnavsky, E., Pierleoni, A., 2018. Evaluation of CHIRPS rainfall estimates over Iran. *Quart J Royal Meteorol Soc* 144, 282–291. <https://doi.org/10.1002/qj.3342>.
- Shen, Z., Yong, B., Gourley, J.J., Qi, W., Lu, D., Liu, J., Ren, L., Hong, Y., Zhang, J., 2020. Recent global performance of the climate hazards group infrared precipitation (CHIRP) with stations (CHIRPS). *J. Hydrol.* 591, 125284. <https://doi.org/10.1016/j.jhydrol.2020.125284>.
- So, D., Shin, D., 2018. Classification of precipitating clouds using satellite infrared observations and its implications for rainfall estimation. *Quart J Royal Meteorol Soc* 144, 133–144. <https://doi.org/10.1002/qj.3288>.
- Su, J., Miao, C., Zwiers, F., Beck, H., Jones, P., Sun, Q., Slater, L.J., Berghuijs, W.R., Wada, Y., Rosenfeld, D., Gou, J., Wu, Y., Tarolli, P., Borrelli, P., Panagos, P., Alexander, L.V., Zhang, Q., Hu, J., Min, S.-K., Samaniego, L., Duan, Q., Destouni, G., Marengo, J.A., Modarres, R., Sorooshian, S., 2026. Precipitation observing network gaps limit climate change impact assessment. *Nature*. <https://doi.org/10.1038/s41586-026-10300-5>.
- Sun, Q., Miao, C., Duan, Q., Ashouri, H., Sorooshian, S., Hsu, K., 2018. A review of global precipitation data sets: data sources, estimation, and intercomparisons. *Rev. Geophys.* 56, 79–107. <https://doi.org/10.1002/2017rg000574>.
- Urrea, V., Ochoa, A., Mesa, O., 2019. Seasonality of rainfall in Colombia. *Water Resour. Res.* 55, 4149–4162. <https://doi.org/10.1029/2018WR023316>.
- Valencia, S., Marín, D.E., Gómez, D., Hoyos, N., Salazar, J.F., Villegas, J.C., 2023a. Spatio-temporal assessment of gridded precipitation products across topographic and climatic gradients in Colombia. *Atmos. Res.* 285, 106643. <https://doi.org/10.1016/j.atmosres.2023.106643>.
- Valencia, S., Salazar, J.F., Hoyos, N., Armenteras, D., Villegas, J.C., 2023b. Current Forest-Savanna transition in northern South America departs from typical climatic thresholds. *Ecosystems*. <https://doi.org/10.1007/s10021-023-00872-y>.
- Vallejo-Bernal, S.M., Urrea, V., Bedoya-Soto, J.M., Posada, D., Olarte, A., Cárdenas-Posso, Y., Ruiz-Murcia, F., Martínez, M.T., Petersen, W.A., Huffman, G.J., Poveda, G., 2021. Ground validation of TRMM 3B43 v7 precipitation estimates over Colombia. Part I: monthly and seasonal timescales. *Int. J. Climatol.* 41, 601–624. <https://doi.org/10.1002/joc.6640>.
- Vargas Godoy, M.R., Markonis, Y., Thomson, J.R., Ballarin, A.S., Perri, S., Miao, C., Sun, Q., Hanel, M., Papalexiou, S.M., Kummerow, C., Oki, T., Molini, A., 2025. Which precipitation dataset to choose for hydrological studies of the terrestrial water cycle? *Bull. Am. Meteorol. Soc.* <https://doi.org/10.1175/BAMS-D-24-0306.1>. BAMS-D-24-0306.1.
- Varma, A.K., 2018. Measurement of precipitation from satellite radiometers (visible, infrared, and microwave): Physical basis, methods, and limitations. In: *Remote sensing of aerosols, clouds, and precipitation*. Elsevier, pp. 223–248. <https://doi.org/10.1016/B978-0-12-810437-8.00011-6>.
- Wang, X., Alharbi, R.S., Baez-Villanueva, O.M., Miralles, D.G., Ma, J., Xu, S., McCabe, M.F., Pappenberger, F., van Dijk, A.I.J.M., McVicar, T.R., Karthikeyan, L., Fowler, H.J., Pan, M., Gebrechorkos, S.H., Beck, H.E., 2026. MSWEP V3: Machine Learning-Powered Global Precipitation Estimates at 0.1° Hourly Resolution (1979–Present). <https://doi.org/10.48550/ARXIV.2602.01436>.
- Woolson, R.F., 2008. Wilcoxon signed-rank test. In: D'Agostino, R.B., Sullivan, L., Massaro, J. (Eds.), *Wiley Encyclopedia of Clinical Trials*. Wiley, pp. 1–3. <https://doi.org/10.1002/9780471462422.eoct979>.
- Xavier, A.C., Scanlon, B.R., King, C.W., Alves, A.I., 2022. New improved Brazilian daily weather gridded data (1961–2020). *Int. J. Climatol.* 42, 8390–8404. <https://doi.org/10.1002/joc.7731>.

- Xie, P., Joyce, R., Wu, S., Yoo, S.-H., Yarosh, Y., Sun, F., Lin, R., 2017. Reprocessed, bias-corrected CMORPH global high-resolution precipitation estimates from 1998. *J. Hydrometeorol.* 18, 1617–1641. <https://doi.org/10.1175/JHM-D-16-0168.1>.
- Xie, Y., Wang, J., Wan, X., Lyu, Y., 2025. Assessing the monthly performance of daily precipitation products over Southeast Asia using the gauge-based analysis. *PLoS One* 20, e0319477. <https://doi.org/10.1371/journal.pone.0319477>.
- Zambrano-Bigiarini, M., Nauditt, A., Birkel, C., Verbist, K., Ribbe, L., 2017. Temporal and spatial evaluation of satellite-based rainfall estimates across the complex topographical and climatic gradients of Chile. *Hydrol. Earth Syst. Sci.* 21, 1295–1320. <https://doi.org/10.5194/hess-21-1295-2017>.
- Zhang, X., Alexander, L., Hegerl, G.C., Jones, P., Tank, A.K., Peterson, T.C., Trewin, B., Zwiers, F.W., 2011. Indices for monitoring changes in extremes based on daily temperature and precipitation data. *WIREs Clim. Change* 2, 851–870. <https://doi.org/10.1002/wcc.147>.